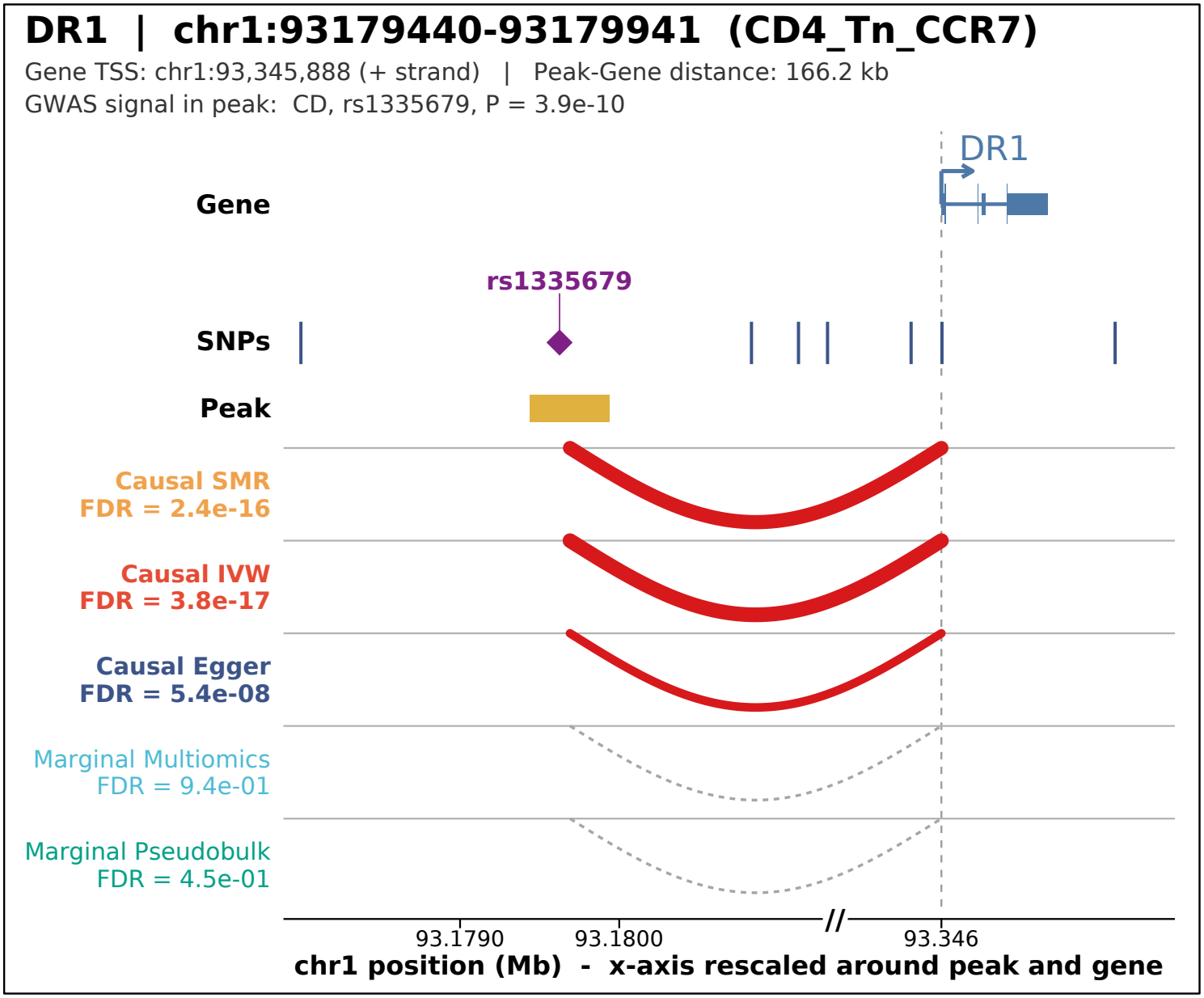
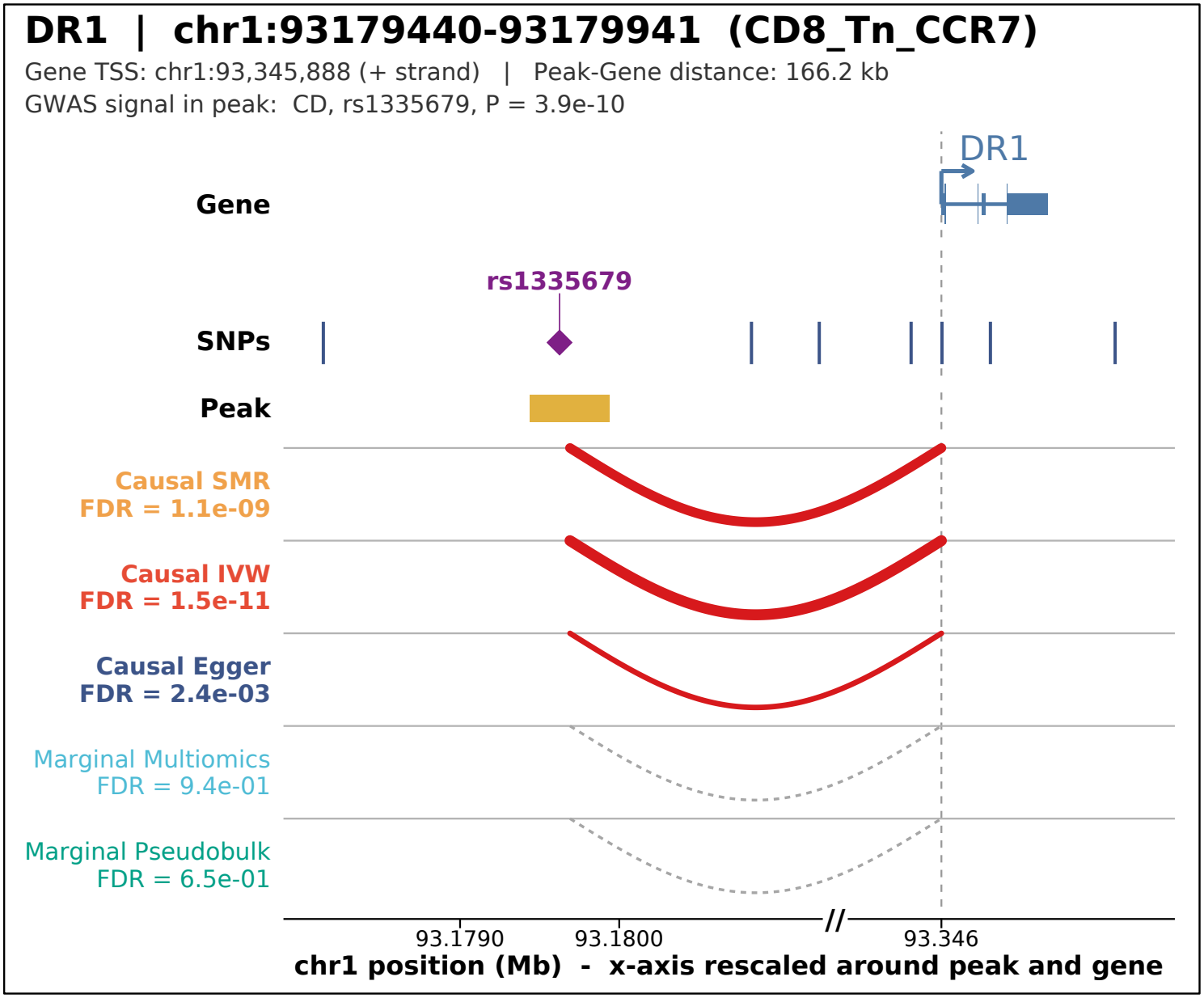
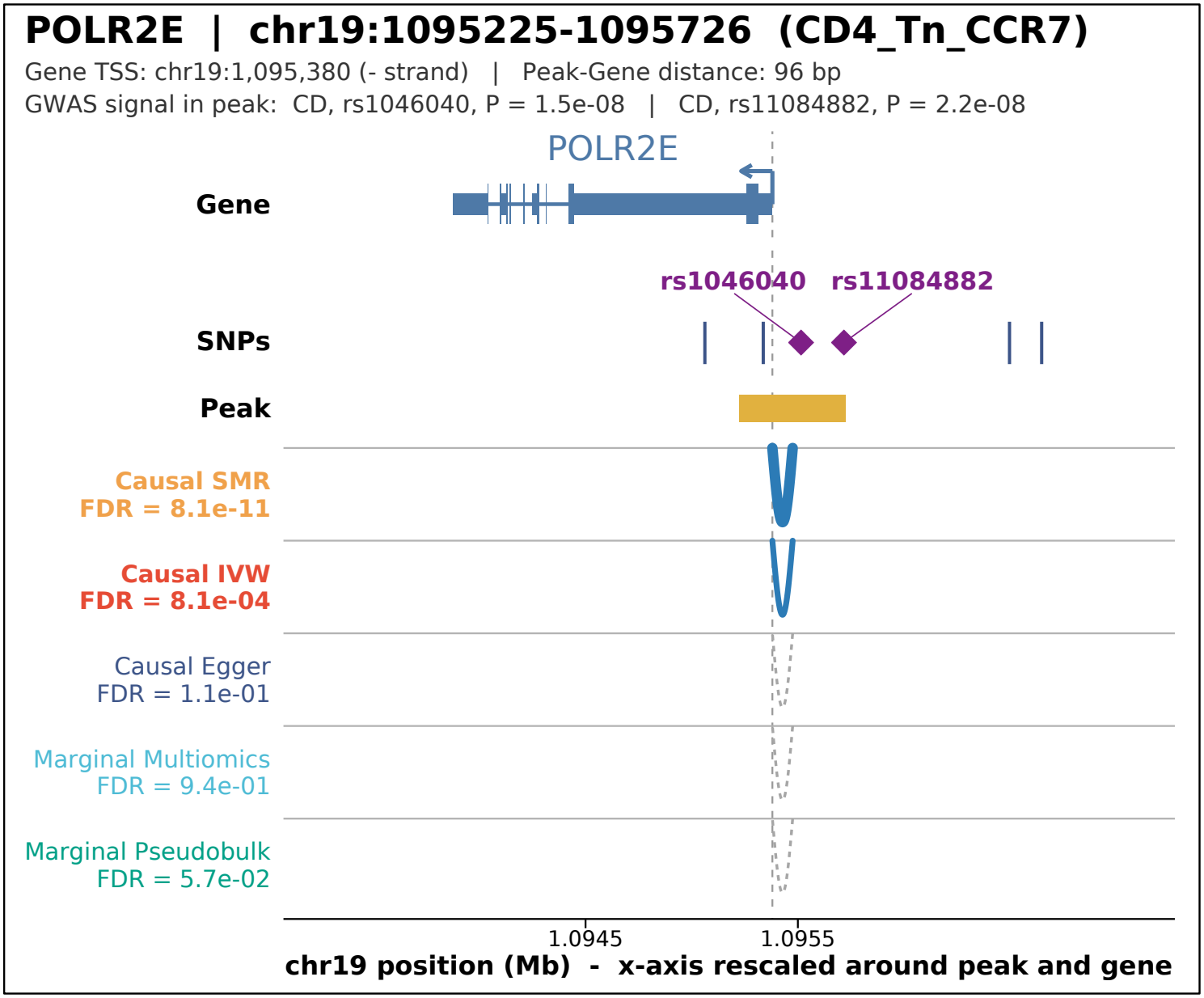
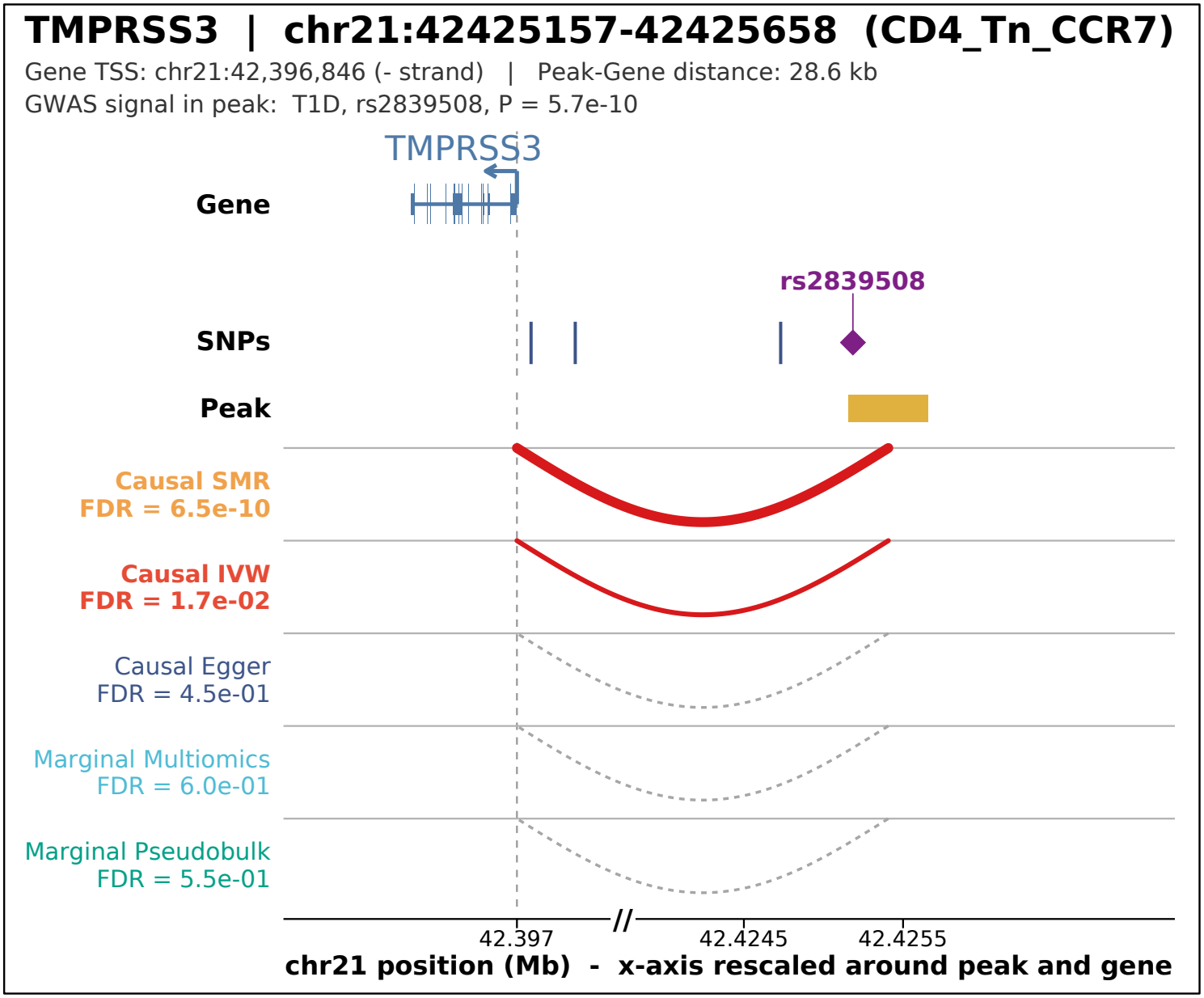
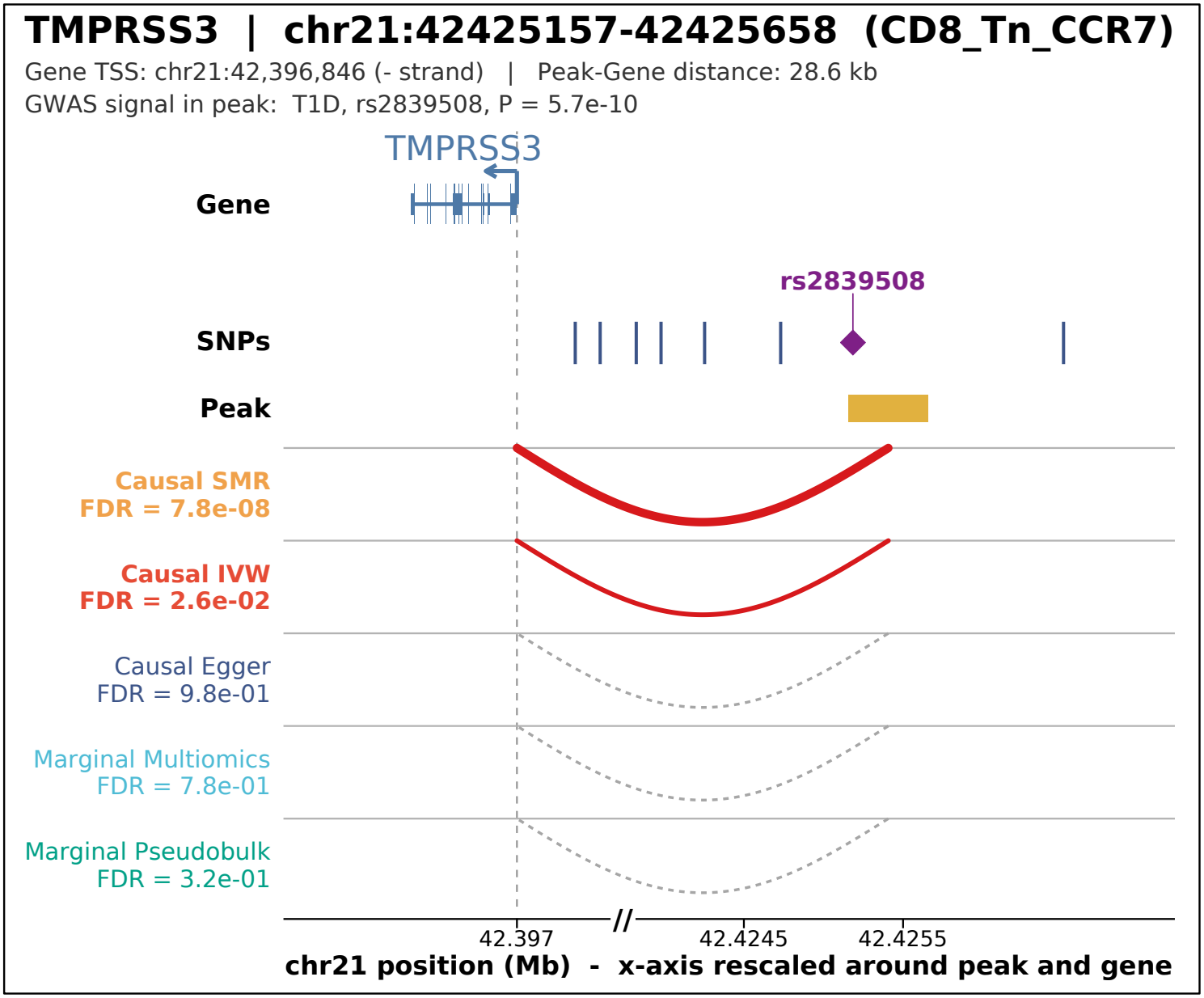
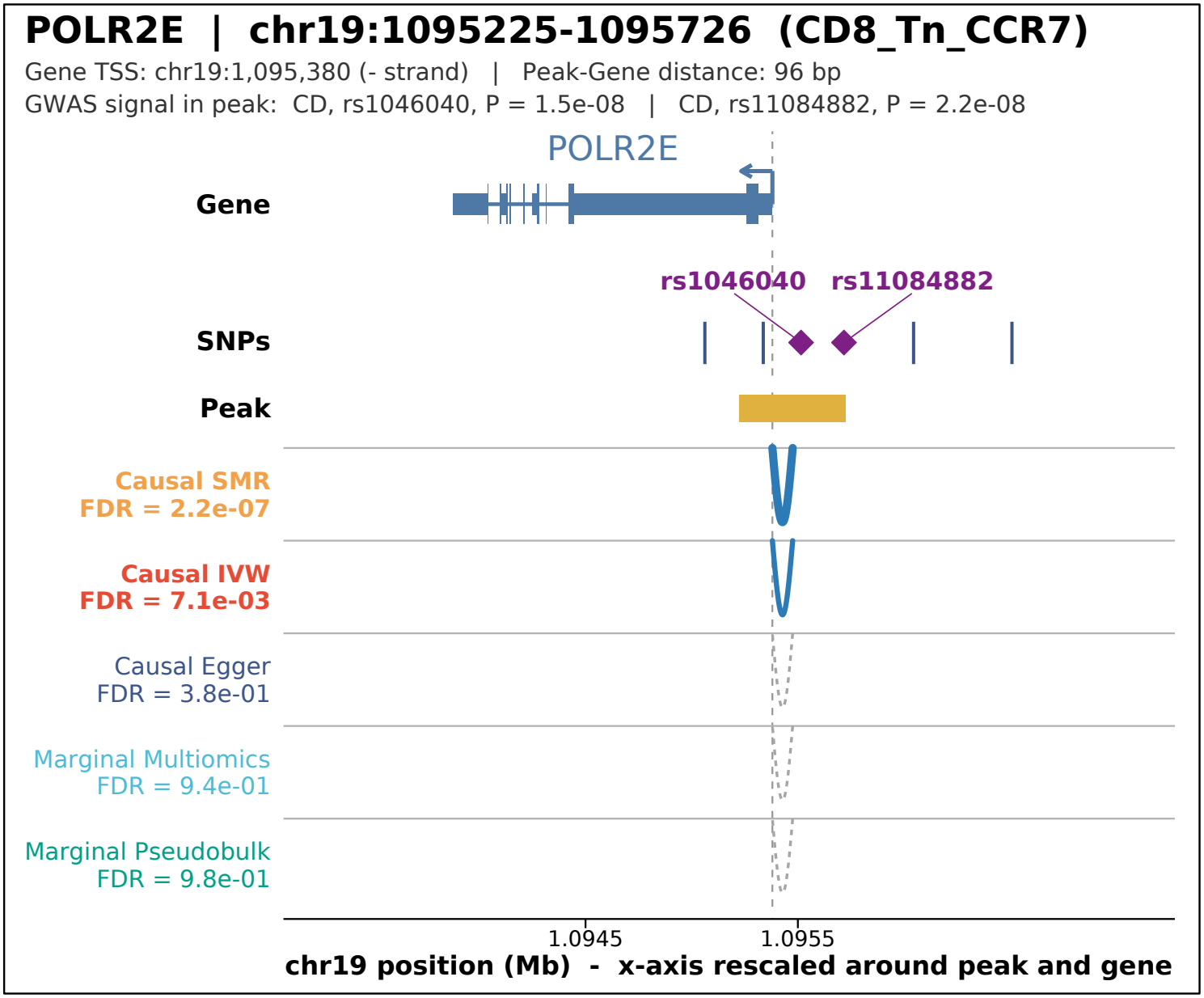

DR1 | chr1:93179440-93179941 (CD4_Tn_CCR7)
Gene TSS: chr1:93,345,888 (+ strand) | Peak-Gene distance: 166.2 kb
GWAS signal in peak: CD, rs1335679, P = 3.9e-10
DR1
Gene
rs1335679
SNPs
Peak
Causal SMR
FDR = 2.4e-16
Causal IVW
FDR = 3.8e-17
Causal Egger
FDR = 5.4e-08
Marginal Multiomics
FDR = 9.4e-01
Marginal Pseudobulk
FDR = 4.5e-01
93.1790
93.1800
93.346
chr1 position (Mb) - x-axis rescaled around peak and gene

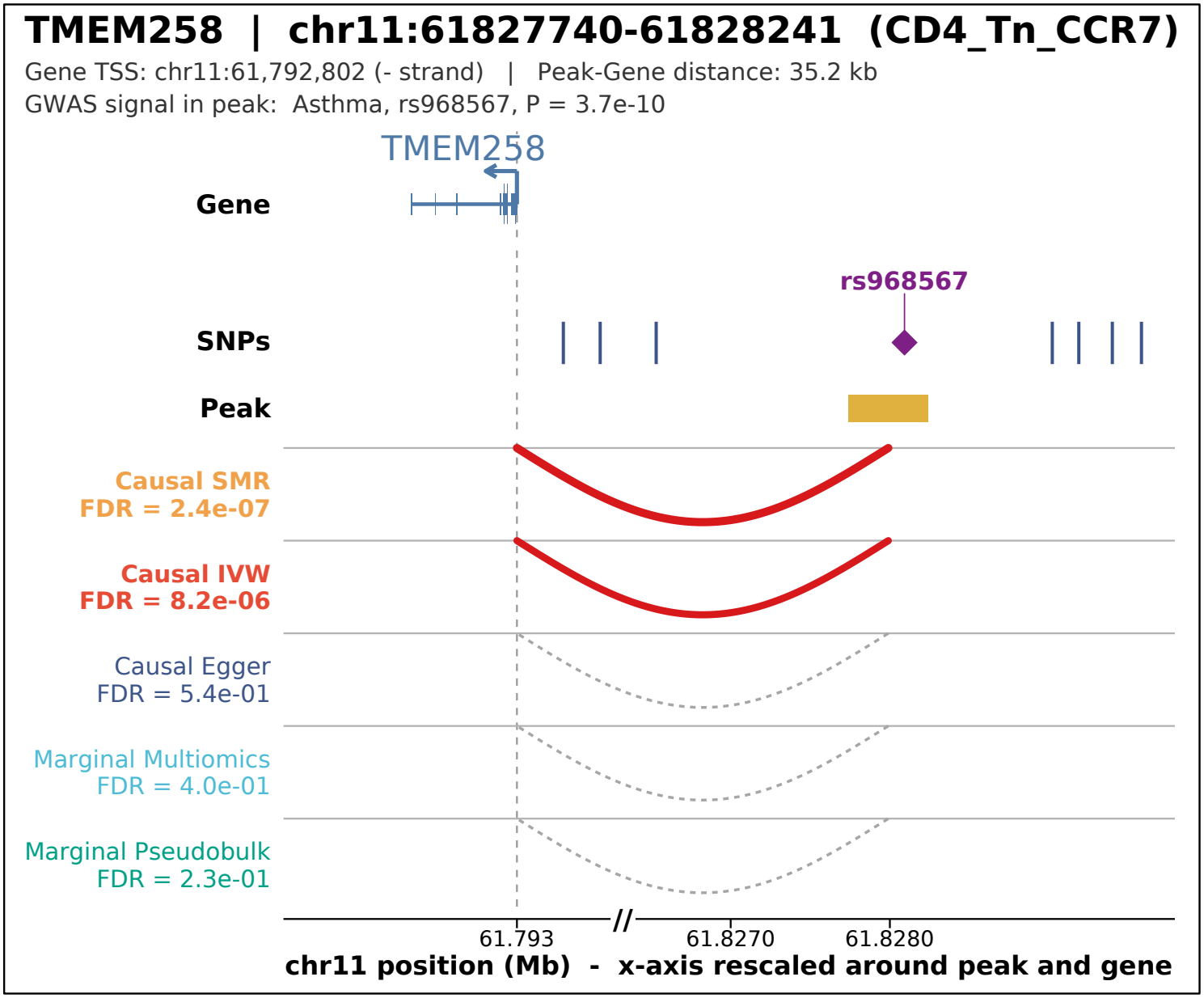

DR1 | chr1:93179440-93179941 (CD8_Tn_CCR7)
Gene TSS: chr1:93,345,888 (+ strand) | Peak-Gene distance: 166.2 kb
GWAS signal in peak: CD, rs1335679, P = 3.9e-10
DR1
Gene
rs1335679
SNPs
Peak
Causal SMR
FDR = 1.1e-09
Causal IVW
FDR = 1.5e-11
Causal Egger
FDR = 2.4e-03
Marginal Multiomics
FDR = 9.4e-01
Marginal Pseudobulk
FDR = 6.5e-01
93.1790
93.1800
93.346
chr1 position (Mb) - x-axis rescaled around peak and gene

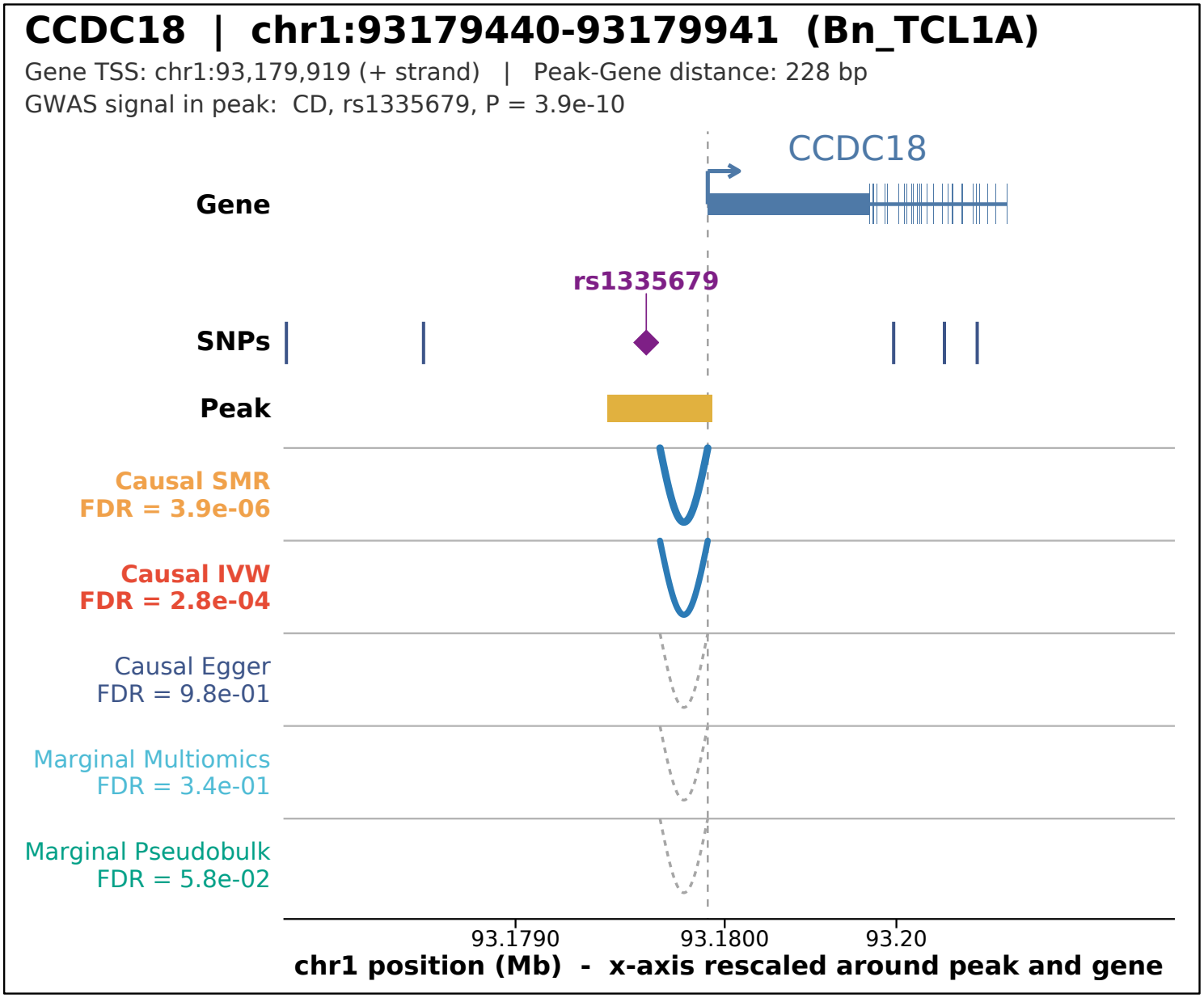

POLR2E | chr19:1095225-1095726 (CD4_Tn_CCR7)
Gene TSS: chr19:1,095,380 (- strand) | Peak-Gene distance: 96 bp
GWAS signal in peak: CD, rs1046040, P = 1.5e-08 | CD, rs11084882, P = 2.2e-08
POLR2E
Gene
rs1046040
rs11084882
SNPs
Peak
Causal SMR
FDR = 8.1e-11
Causal IVW
FDR = 8.1e-04
Causal Egger
FDR = 1.1e-01
Marginal Multiomics
FDR = 9.4e-01
Marginal Pseudobulk
FDR = 5.7e-02
1.0945
1.0955
chr19 position (Mb) - x-axis rescaled around peak and gene

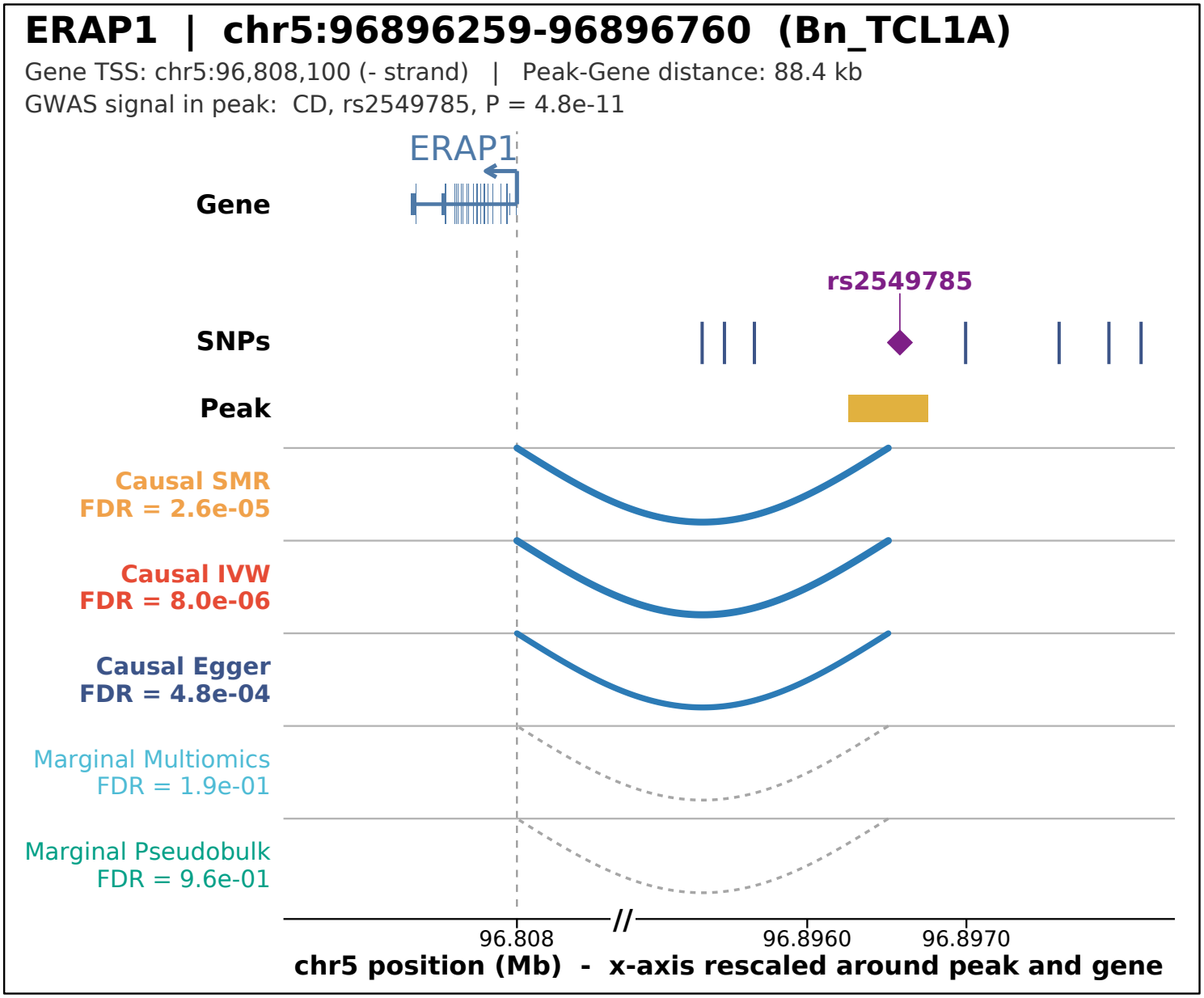

TMPRSS3 | chr21:42425157-42425658 (CD4_Tn_CCR7)
Gene TSS: chr21:42,396,846 (- strand) | Peak-Gene distance: 28.6 kb
GWAS signal in peak: T1D, rs2839508, P = 5.7e-10
TMPRSS3
Gene
rs2839508
SNPs
Peak
Causal SMR
FDR = 6.5e-10
Causal IVW
FDR = 1.7e-02
Causal Egger
FDR = 4.5e-01
Marginal Multiomics
FDR = 6.0e-01
Marginal Pseudobulk
FDR = 5.5e-01
42.397
42.4245
42.4255
chr21 position (Mb) - x-axis rescaled around peak and gene

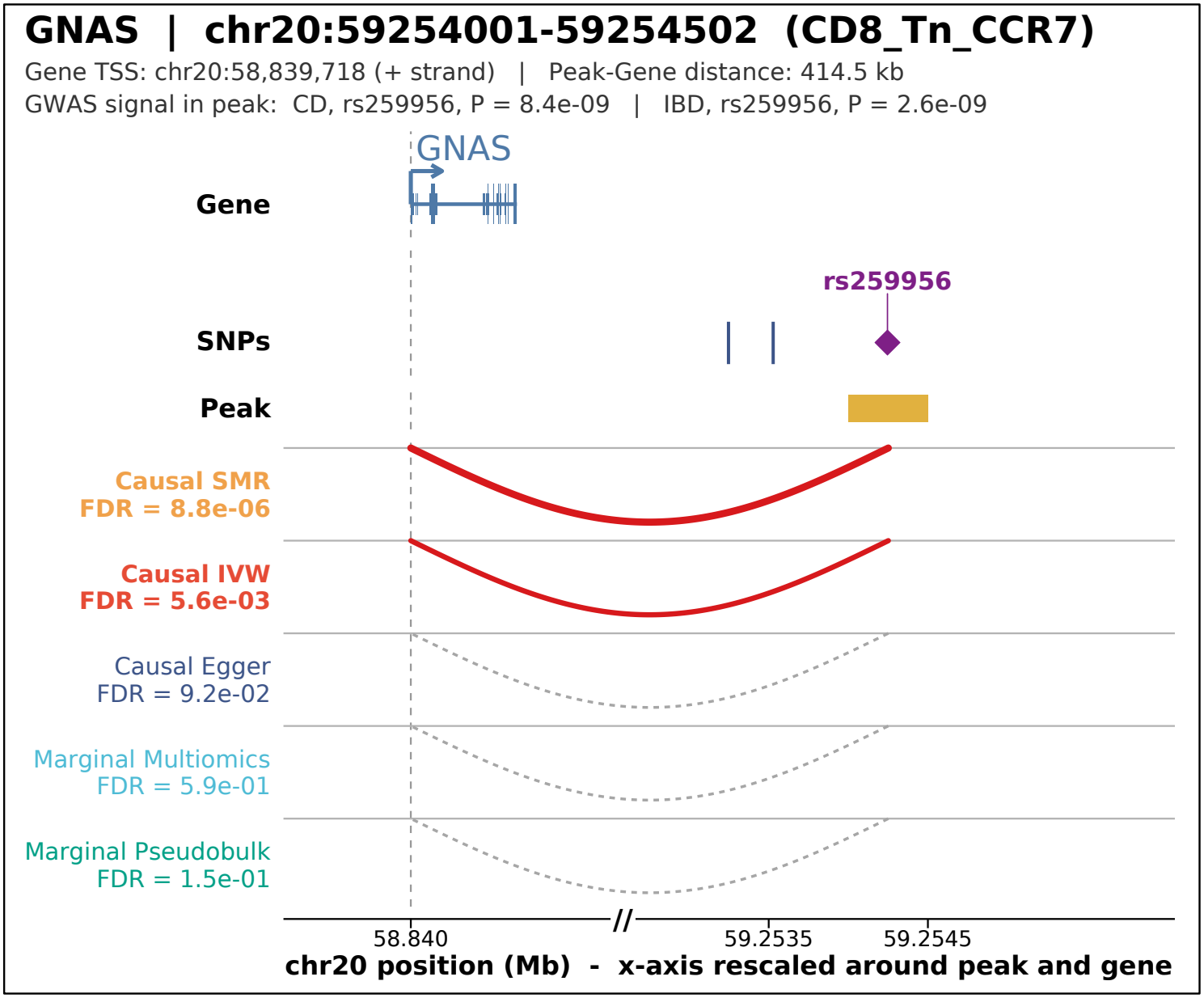

TMPRSS3 | chr21:42425157-42425658 (CD8_Tn_CCR7)
Gene TSS: chr21:42,396,846 (- strand) | Peak-Gene distance: 28.6 kb
GWAS signal in peak: T1D, rs2839508, P = 5.7e-10
TMPRSS3
Gene
rs2839508
SNPs
Peak
Causal SMR
FDR = 7.8e-08
Causal IVW
FDR = 2.6e-02
Causal Egger
FDR = 9.8e-01
Marginal Multiomics
FDR = 7.8e-01
Marginal Pseudobulk
FDR = 3.2e-01
42.397
42.4245
42.4255
chr21 position (Mb) - x-axis rescaled around peak and gene

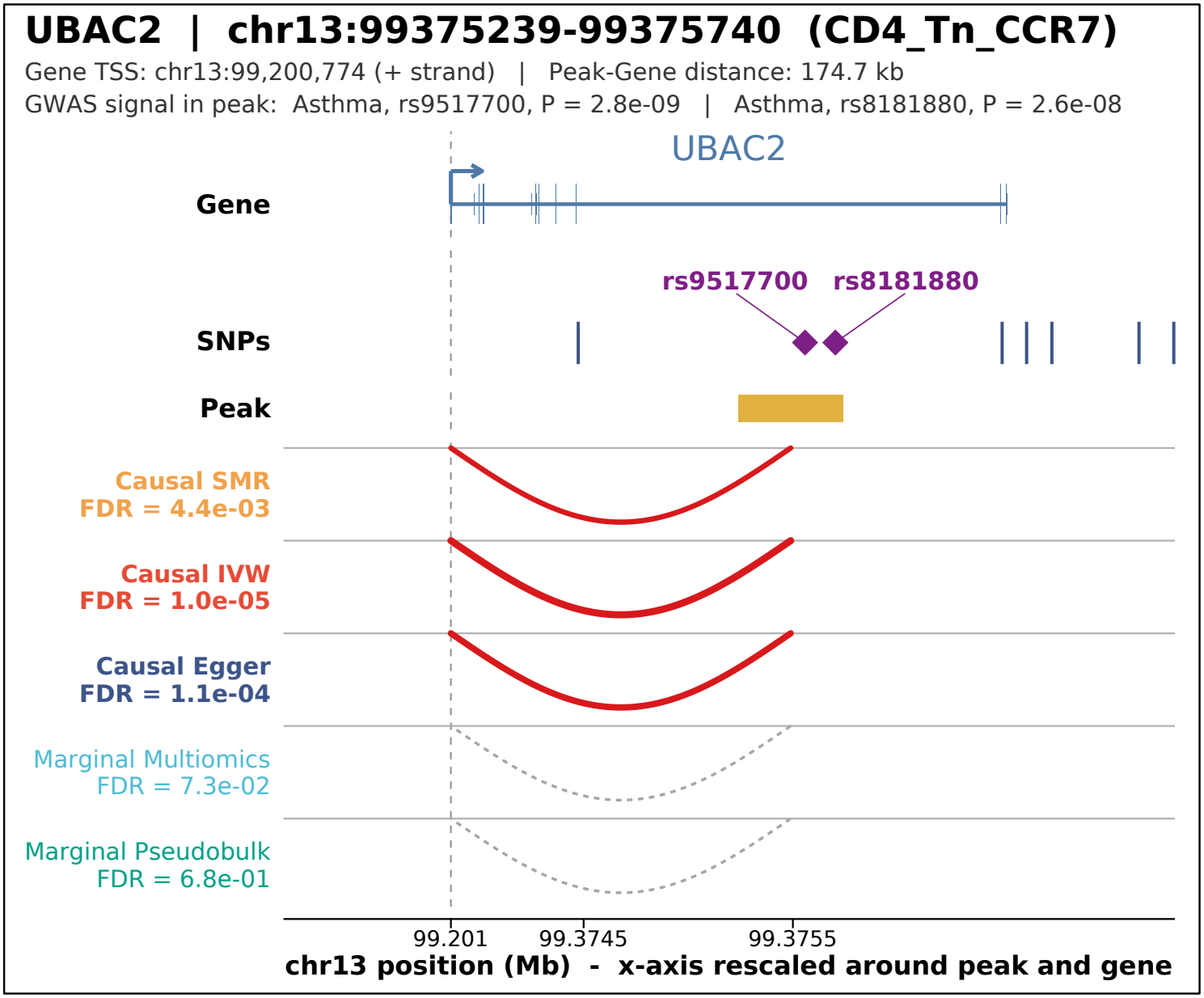

POLR2E | chr19:1095225-1095726 (CD8_Tn_CCR7)
Gene TSS: chr19:1,095,380 (- strand) | Peak-Gene distance: 96 bp
GWAS signal in peak: CD, rs1046040, P = 1.5e-08 | CD, rs11084882, P = 2.2e-08
POLR2E
Gene
rs1046040
rs11084882
SNPs
Peak
Causal SMR
FDR = 2.2e-07
Causal IVW
FDR = 7.1e-03
Causal Egger
FDR = 3.8e-01
Marginal Multiomics
FDR = 9.4e-01
Marginal Pseudobulk
FDR = 9.8e-01
1.0945
1.0955
chr19 position (Mb) - x-axis rescaled around peak and gene

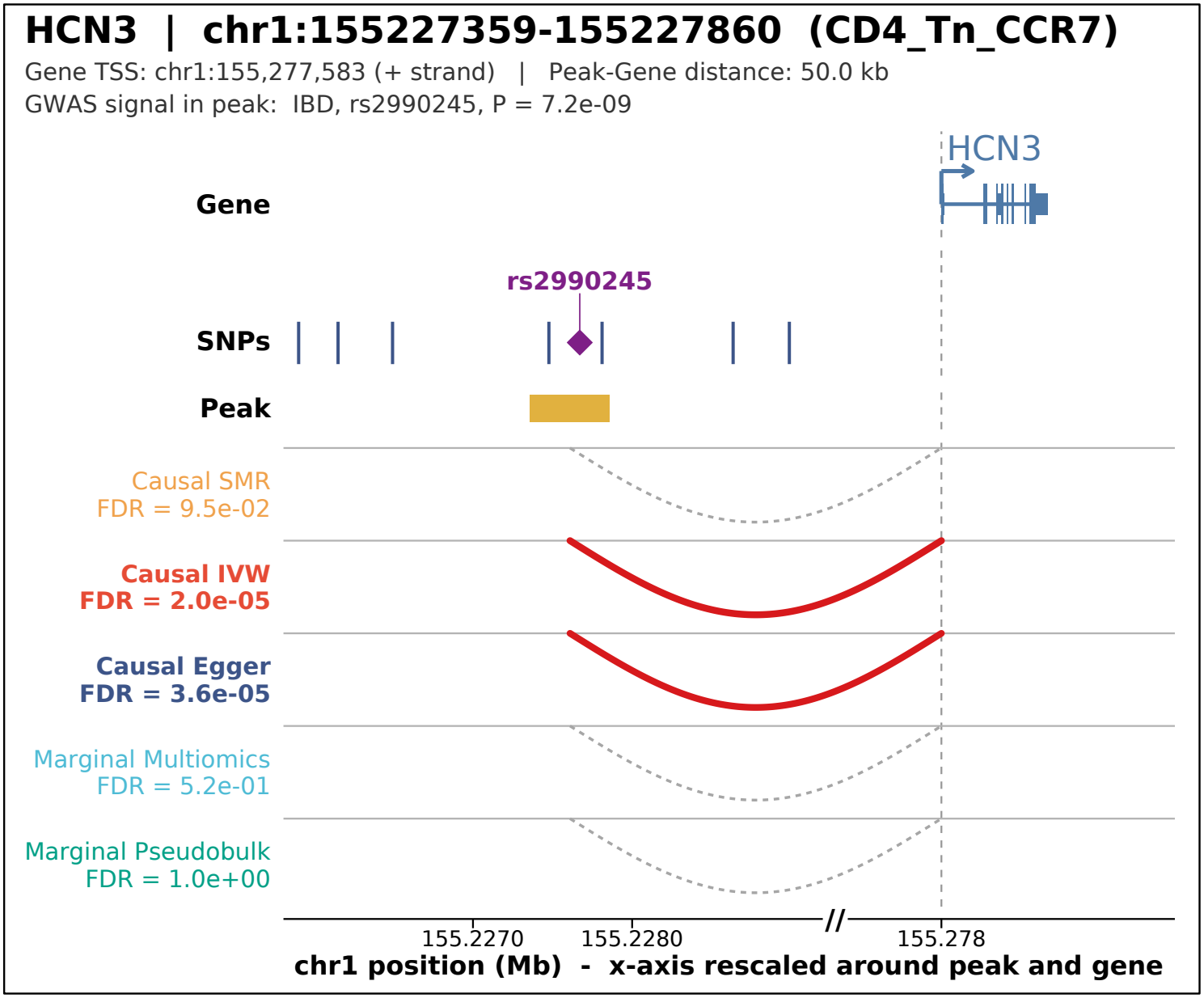

TMEM258 | chr11:61827740-61828241 (CD4_Tn_CCR7)
Gene TSS: chr11:61,792,802 (- strand) | Peak-Gene distance: 35.2 kb
GWAS signal in peak: Asthma, rs968567, P = 3.7e-10
TMEM258
Gene
rs968567
SNPs
Peak
Causal SMR
FDR = 2.4e-07
Causal IVW
FDR = 8.2e-06
Causal Egger
FDR = 5.4e-01
Marginal Multiomics
FDR = 4.0e-01
Marginal Pseudobulk
FDR = 2.3e-01
61.793
61.8270
61.8280
chr11 position (Mb) - x-axis rescaled around peak and gene

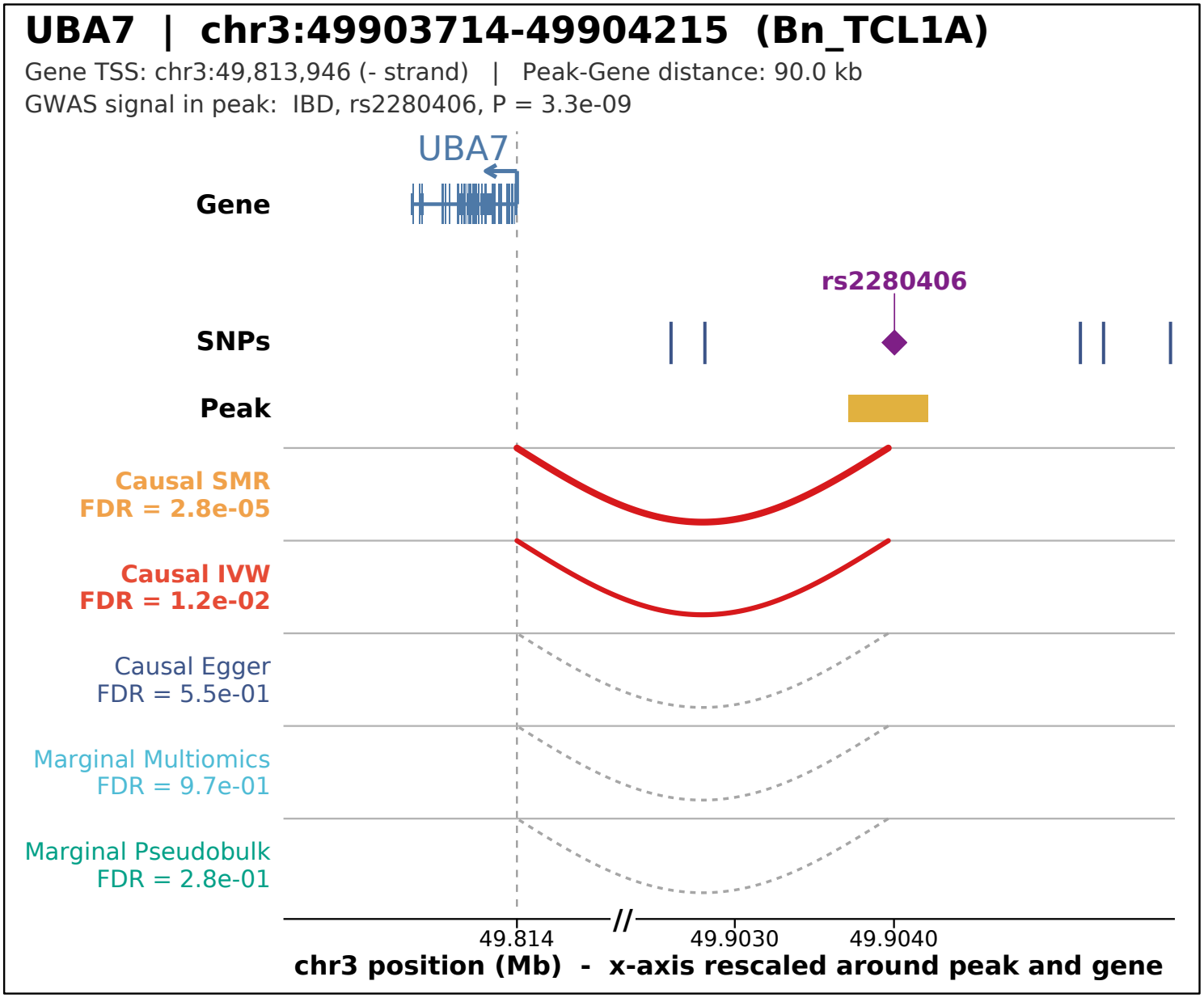

CCDC18 | chr1:93179440-93179941 (Bn_TCL1A)
Gene TSS: chr1:93,179,919 (+ strand) | Peak-Gene distance: 228 bp
GWAS signal in peak: CD, rs1335679, P = 3.9e-10
CCDC18
Gene
rs1335679
SNPs
Peak
Causal SMR
FDR = 3.9e-06
Causal IVW
FDR = 2.8e-04
Causal Egger
FDR = 9.8e-01
Marginal Multiomics
FDR = 3.4e-01
Marginal Pseudobulk
FDR = 5.8e-02
93.1790
93.1800
93.20
chr1 position (Mb) - x-axis rescaled around peak and gene

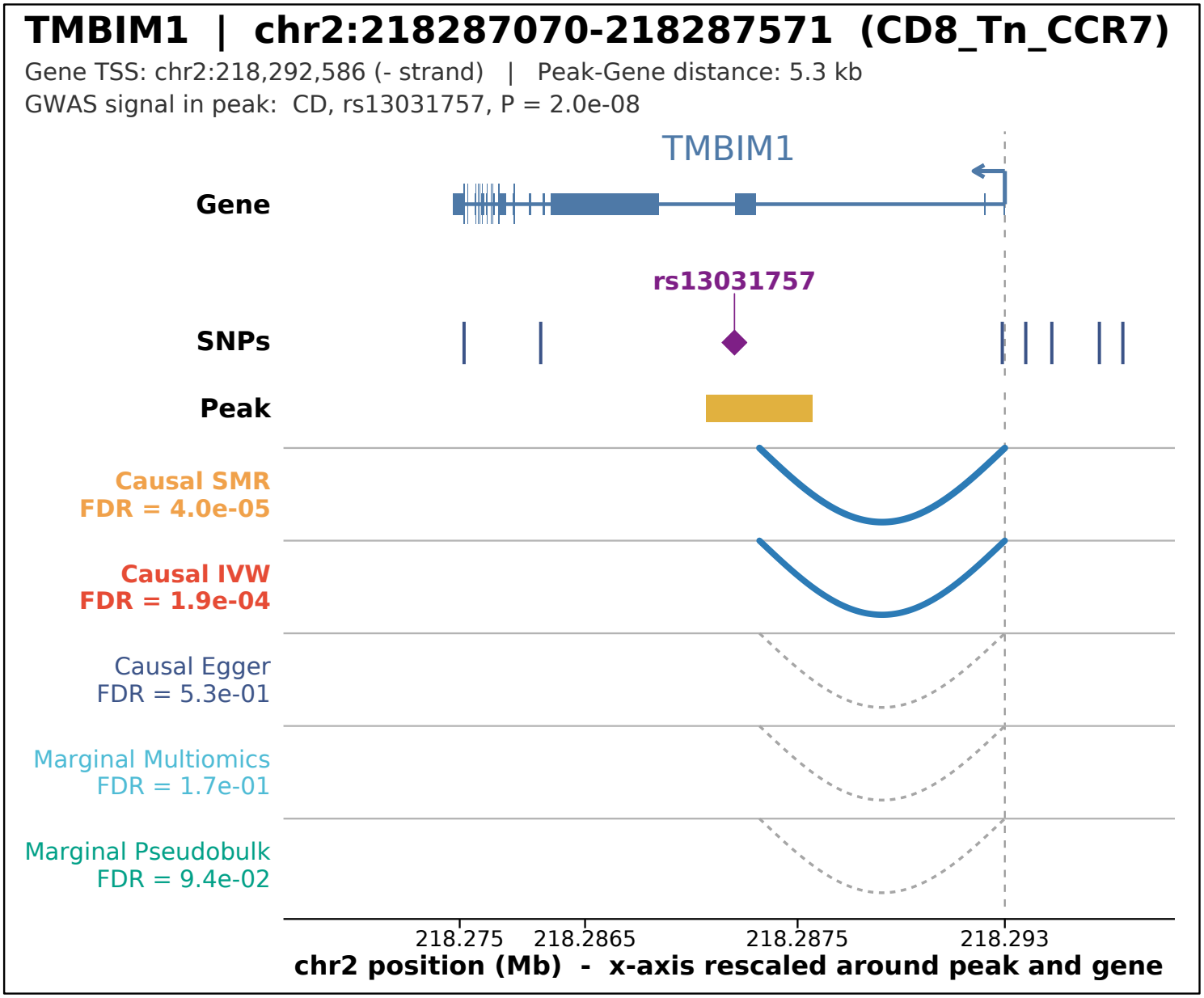

ERAP1 | chr5:96896259-96896760 (Bn_TCL1A)
Gene TSS: chr5:96,808,100 (- strand) | Peak-Gene distance: 88.4 kb
GWAS signal in peak: CD, rs2549785, P = 4.8e-11
ERAP1
Gene
rs2549785
SNPs
Peak
Causal SMR
FDR = 2.6e-05
Causal IVW
FDR = 8.0e-06
Causal Egger
FDR = 4.8e-04
Marginal Multiomics
FDR = 1.9e-01
Marginal Pseudobulk
FDR = 9.6e-01
96.808
//
96.8960
96.8970
chr5 position (Mb) - x-axis rescaled around peak and gene

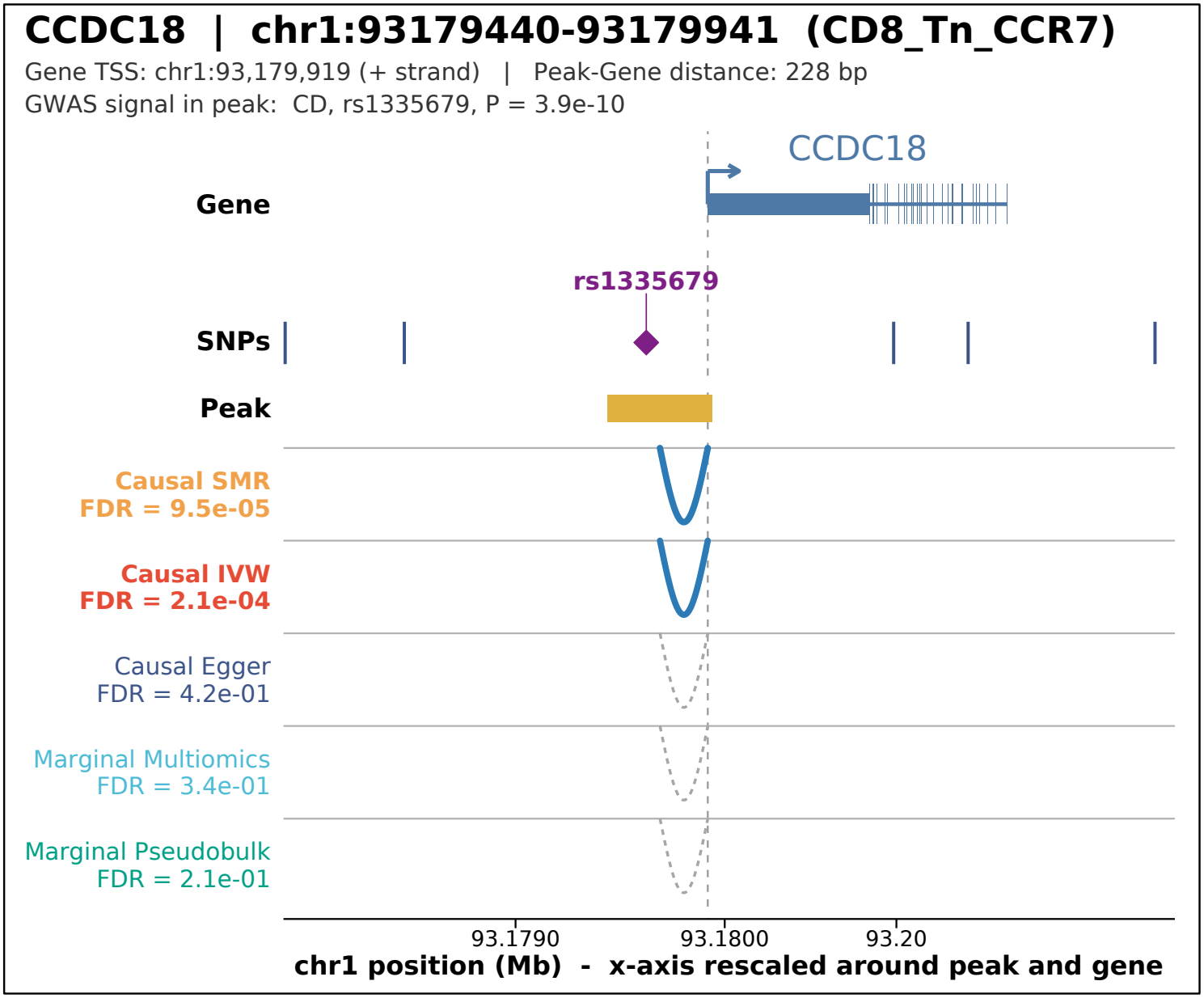

GNAS | chr20:59254001-59254502 (CD8_Tn_CCR7)
Gene TSS: chr20:58,839,718 (+ strand) | Peak-Gene distance: 414.5 kb
GWAS signal in peak: CD, rs259956, P = 8.4e-09 | IBD, rs259956, P = 2.6e-09
GNAS
Gene
rs259956
SNPs
Peak
Causal SMR
FDR = 8.8e-06
Causal IVW
FDR = 5.6e-03
Causal Egger
FDR = 9.2e-02
Marginal Multiomics
FDR = 5.9e-01
Marginal Pseudobulk
FDR = 1.5e-01
58.840
59.2535
59.2545
chr20 position (Mb) - x-axis rescaled around peak and gene

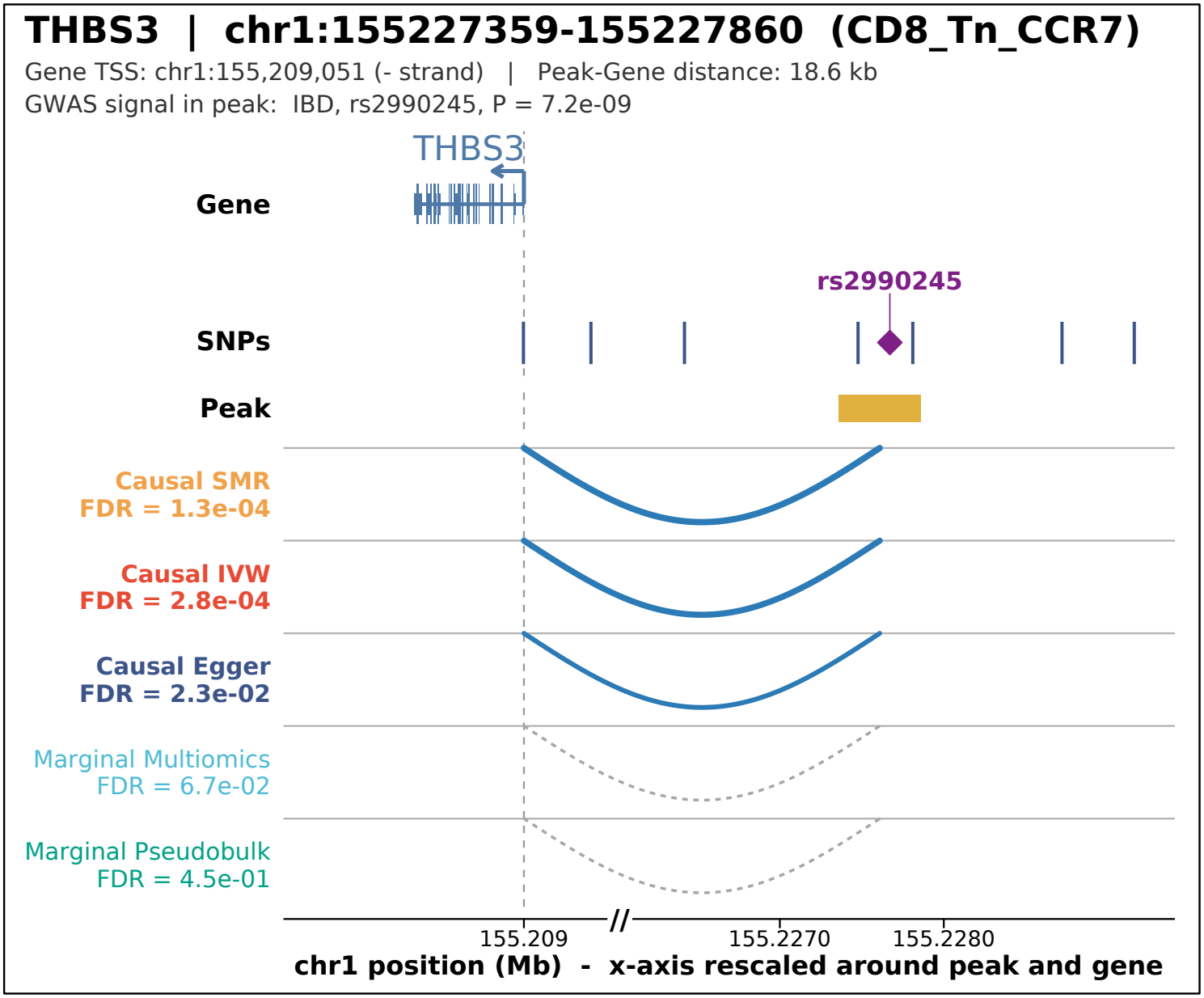

UBAC2 | chr13:99375239-99375740 (CD4_Tn_CCR7)
Gene TSS: chr13:99,200,774 (+ strand) | Peak-Gene distance: 174.7 kb
GWAS signal in peak: Asthma, rs9517700, P = 2.8e-09 | Asthma, rs8181880, P = 2.6e-08
UBAC2
Gene
rs9517700
rs8181880
SNPs
Peak
Causal SMR
FDR = 4.4e-03
Causal IVW
FDR = 1.0e-05
Causal Egger
FDR = 1.1e-04
Marginal Multiomics
FDR = 7.3e-02
Marginal Pseudobulk
FDR = 6.8e-01
99.201
99.3745
99.3755
chr13 position (Mb) - x-axis rescaled around peak and gene

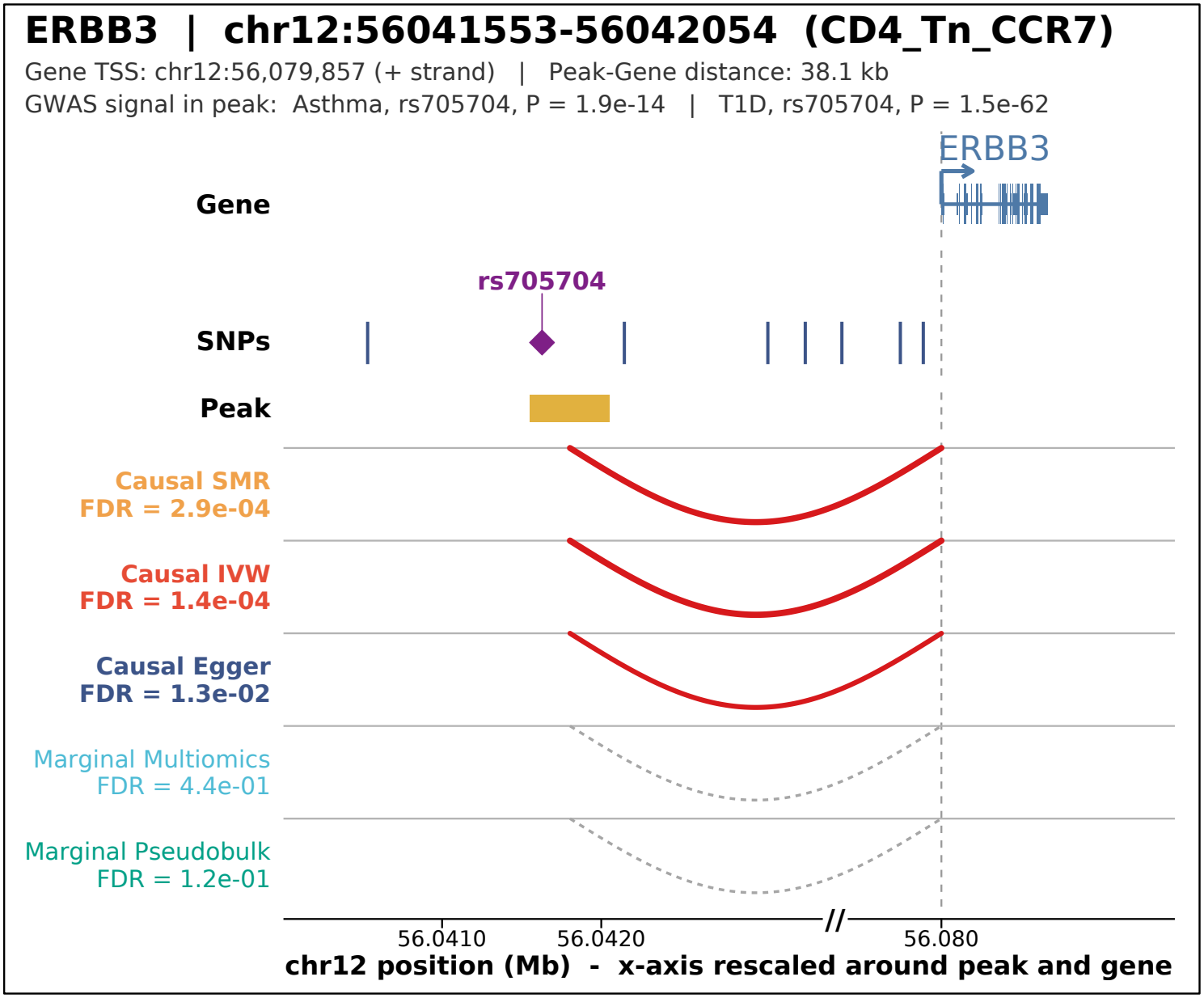

HCN3 | chr1:155227359-155227860 (CD4_Tn_CCR7)
Gene TSS: chr1:155,277,583 (+ strand) | Peak-Gene distance: 50.0 kb
GWAS signal in peak: IBD, rs2990245, P = 7.2e-09
HCN3
Gene
rs2990245
SNPs
Peak
Causal SMR
FDR = 9.5e-02
Causal IVW
FDR = 2.0e-05
Causal Egger
FDR = 3.6e-05
Marginal Multiomics
FDR = 5.2e-01
Marginal Pseudobulk
FDR = 1.0e+00
155.2270
155.2280
155.278
chr1 position (Mb) - x-axis rescaled around peak and gene

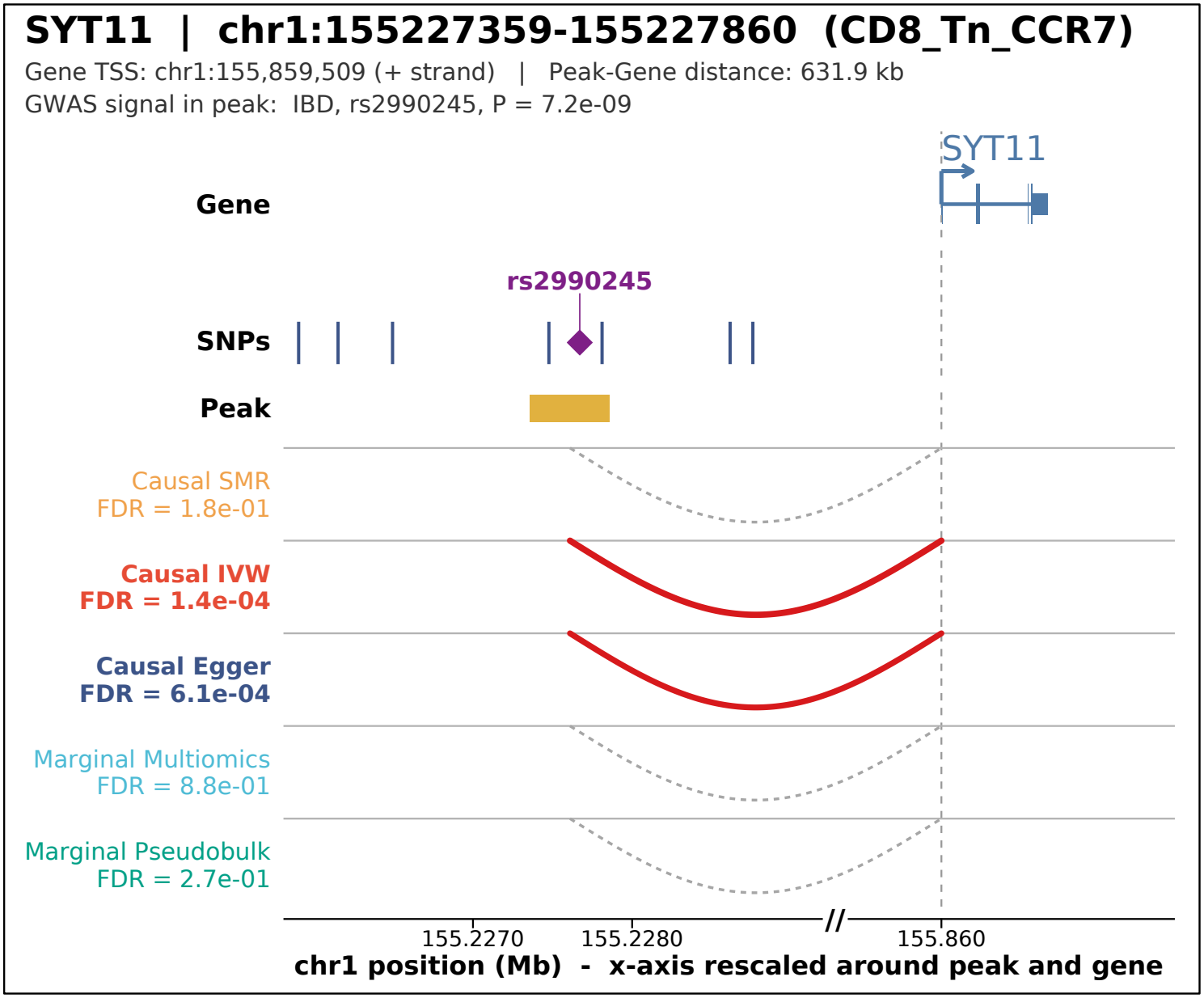

UBA7 | chr3:49903714-49904215 (Bn_TCL1A)
Gene TSS: chr3:49,813,946 (- strand) | Peak-Gene distance: 90.0 kb
GWAS signal in peak: IBD, rs2280406, P = 3.3e-09
UBA7
Gene
rs2280406
SNPs
Peak
Causal SMR
FDR = 2.8e-05
Causal IVW
FDR = 1.2e-02
Causal Egger
FDR = 5.5e-01
Marginal Multiomics
FDR = 9.7e-01
Marginal Pseudobulk
FDR = 2.8e-01
49.814
49.9030
49.9040
chr3 position (Mb) - x-axis rescaled around peak and gene

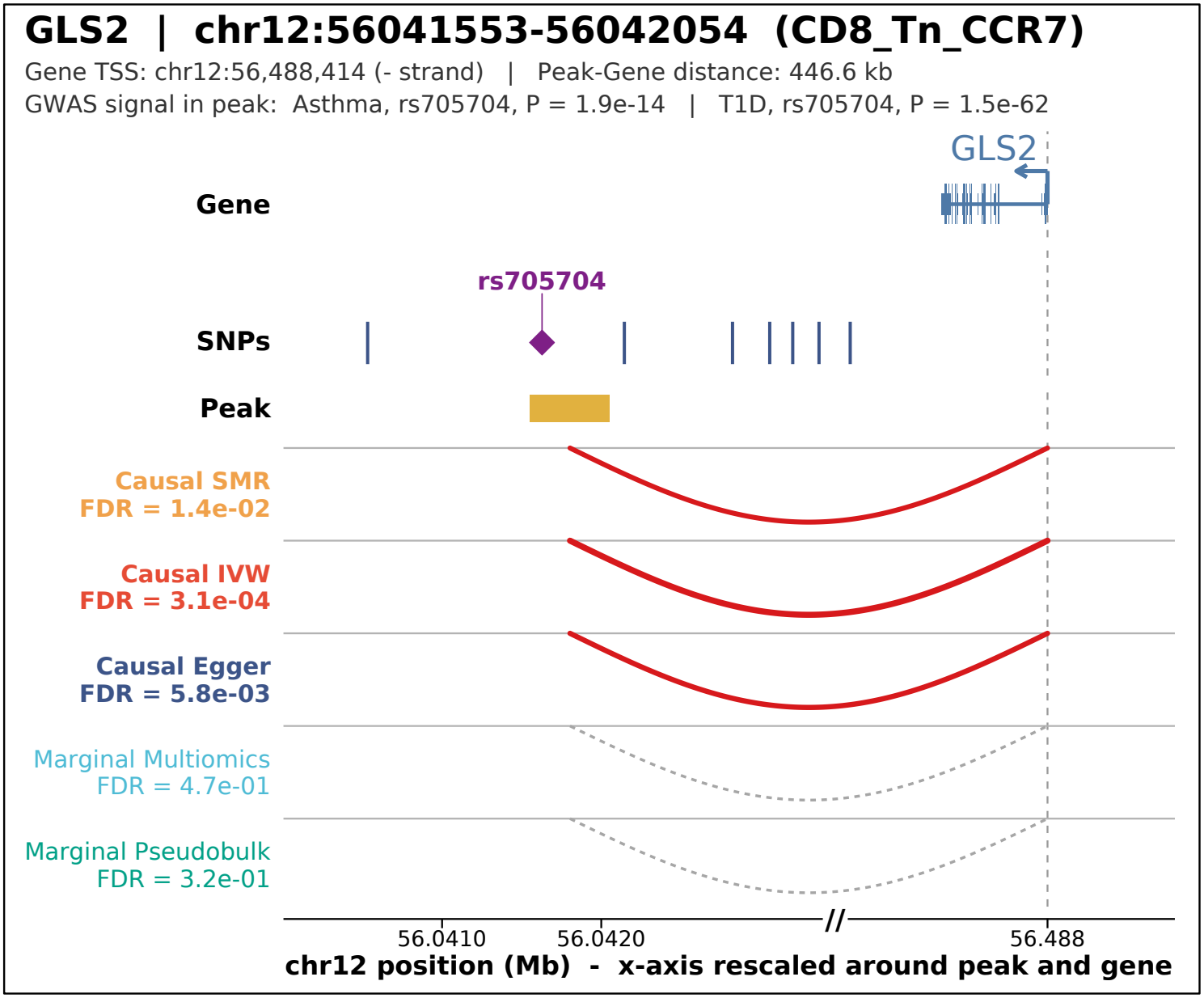

TMBIM1 | chr2:218287070-218287571 (CD8_Tn_CCR7)
Gene TSS: chr2:218,292,586 (- strand) | Peak-Gene distance: 5.3 kb
GWAS signal in peak: CD, rs13031757, P = 2.0e-08
TMBIM1
Gene
rs13031757
SNPs
Peak
Causal SMR
FDR = 4.0e-05
Causal IVW
FDR = 1.9e-04
Causal Egger
FDR = 5.3e-01
Marginal Multiomics
FDR = 1.7e-01
Marginal Pseudobulk
FDR = 9.4e-02
218.275
218.2865
218.2875
218.293
chr2 position (Mb) - x-axis rescaled around peak and gene

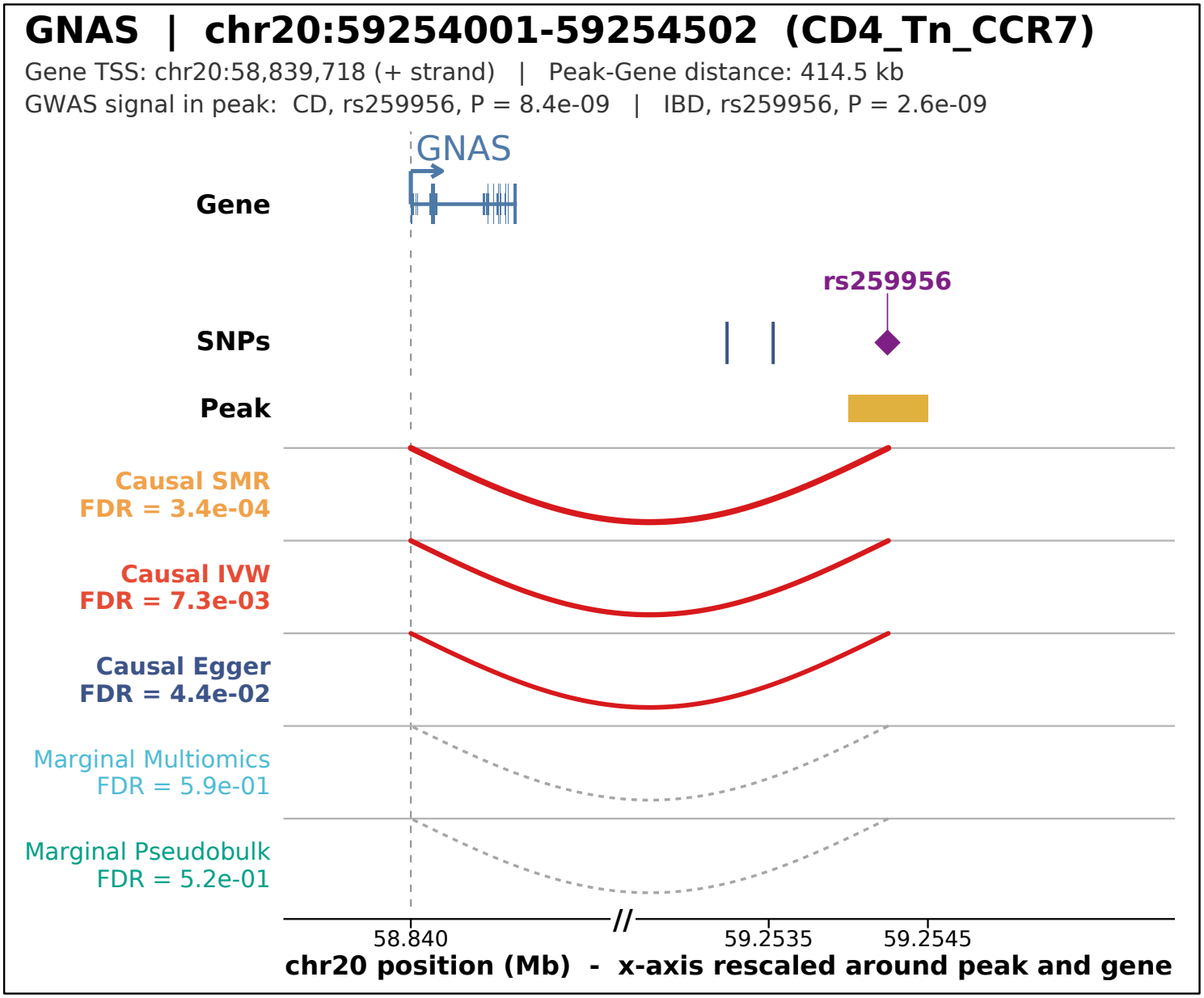

CCDC18 | chr1:93179440-93179941 (CD8_Tn_CCR7)
Gene TSS: chr1:93,179,919 (+ strand) | Peak-Gene distance: 228 bp
GWAS signal in peak: CD, rs1335679, P = 3.9e-10
CCDC18
Gene
rs1335679
SNPs
Peak
Causal SMR
FDR = 9.5e-05
Causal IVW
FDR = 2.1e-04
Causal Egger
FDR = 4.2e-01
Marginal Multiomics
FDR = 3.4e-01
Marginal Pseudobulk
FDR = 2.1e-01
93.1790
93.1800
93.20
chr1 position (Mb) - x-axis rescaled around peak and gene

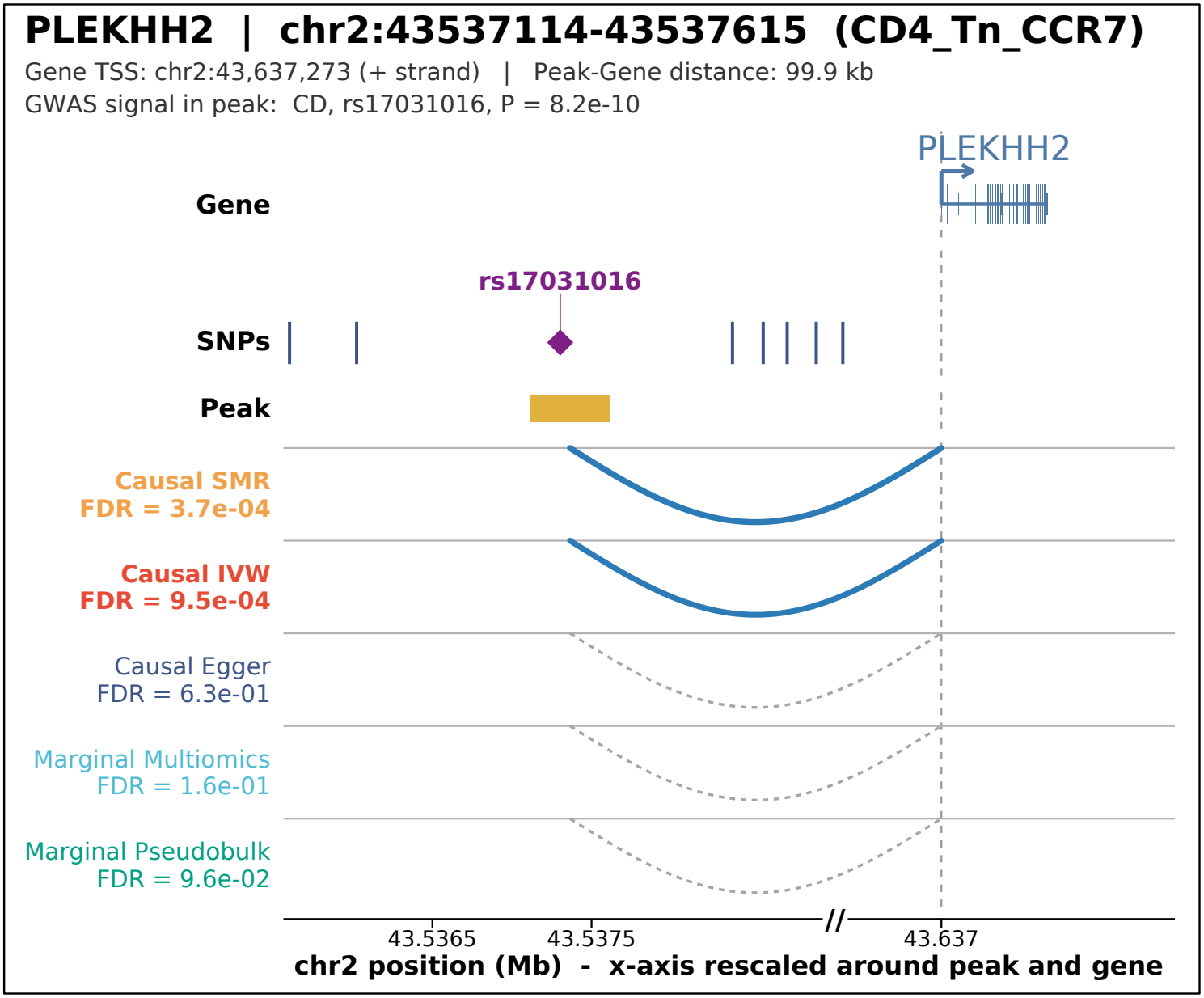

THBS3 | chr1:155227359-155227860 (CD8_Tn_CCR7)
Gene TSS: chr1:155,209,051 (- strand) | Peak-Gene distance: 18.6 kb
GWAS signal in peak: IBD, rs2990245, P = 7.2e-09
THBS3
Gene
rs2990245
SNPs
Peak
Causal SMR
FDR = 1.3e-04
Causal IVW
FDR = 2.8e-04
Causal Egger
FDR = 2.3e-02
Marginal Multiomics
FDR = 6.7e-02
Marginal Pseudobulk
FDR = 4.5e-01
155.209
155.2270
155.2280
chr1 position (Mb) - x-axis rescaled around peak and gene

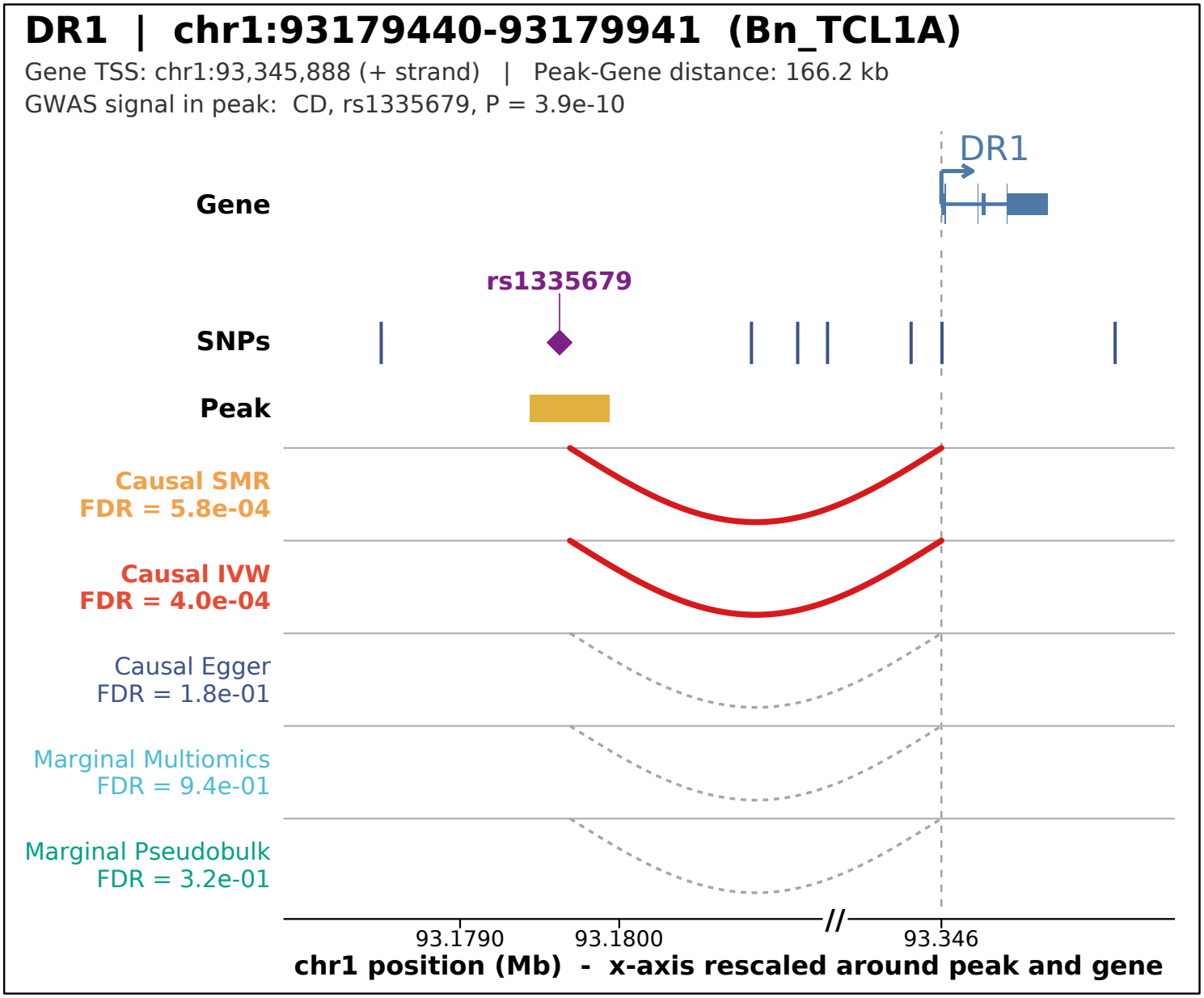

ERBB3 | chr12:56041553-56042054 (CD4_Tn_CCR7)
Gene TSS: chr12:56,079,857 (+ strand) | Peak-Gene distance: 38.1 kb
GWAS signal in peak: Asthma, rs705704, P = 1.9e-14 | T1D, rs705704, P = 1.5e-62
ERBB3
Gene
rs705704
SNPs
Peak
Causal SMR
FDR = 2.9e-04
Causal IVW
FDR = 1.4e-04
Causal Egger
FDR = 1.3e-02
Marginal Multiomics
FDR = 4.4e-01
Marginal Pseudobulk
FDR = 1.2e-01
56.0410
56.0420
56.080
chr12 position (Mb) - x-axis rescaled around peak and gene

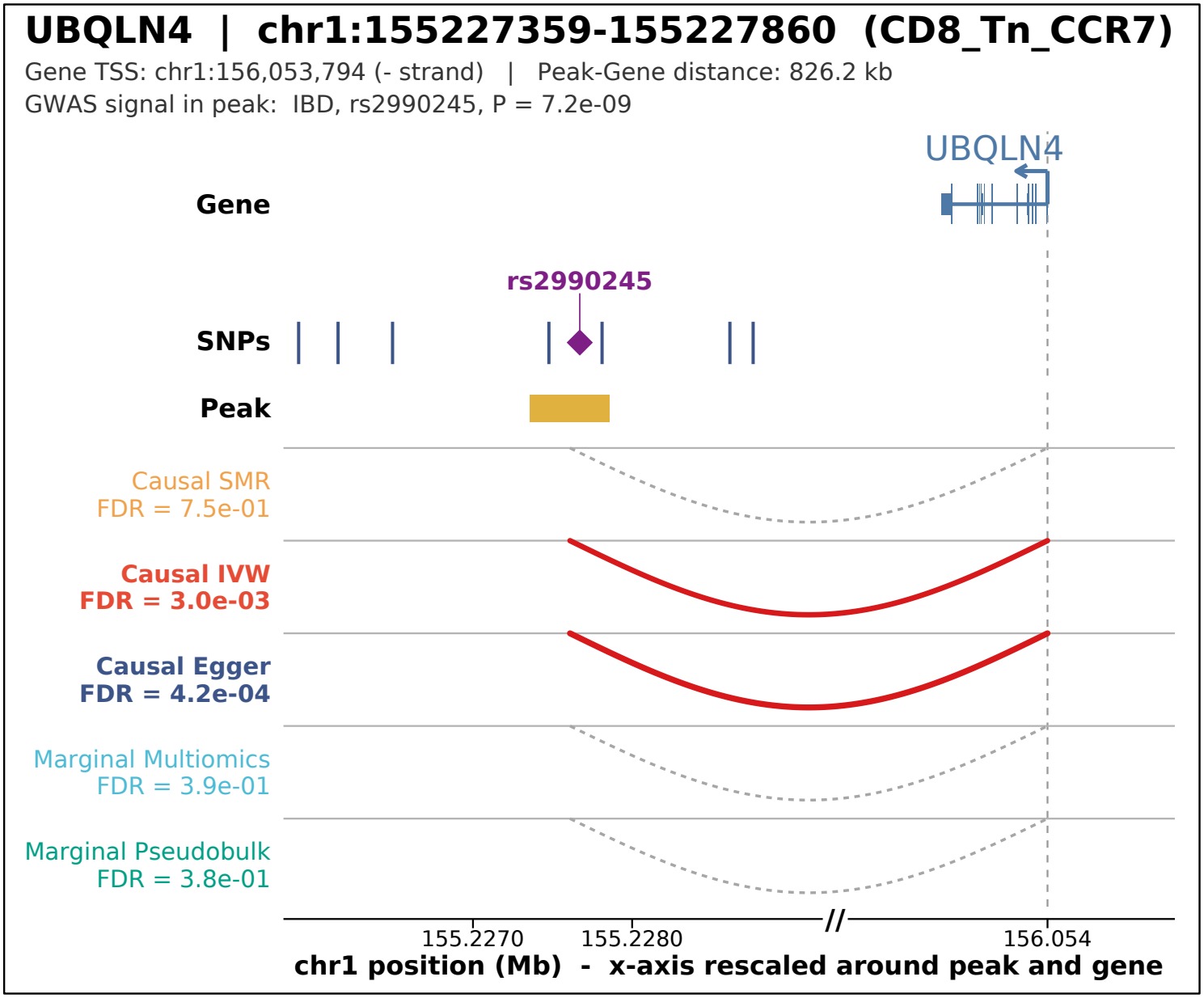

SYT11 | chr1:155227359-155227860 (CD8_Tn_CCR7)
Gene TSS: chr1:155,859,509 (+ strand) | Peak-Gene distance: 631.9 kb
GWAS signal in peak: IBD, rs2990245, P = 7.2e-09
SYT11
Gene
rs2990245
SNPs
Peak
Causal SMR
FDR = 1.8e-01
Causal IVW
FDR = 1.4e-04
Causal Egger
FDR = 6.1e-04
Marginal Multiomics
FDR = 8.8e-01
Marginal Pseudobulk
FDR = 2.7e-01
155.2270
155.2280
155.860
chr1 position (Mb) - x-axis rescaled around peak and gene

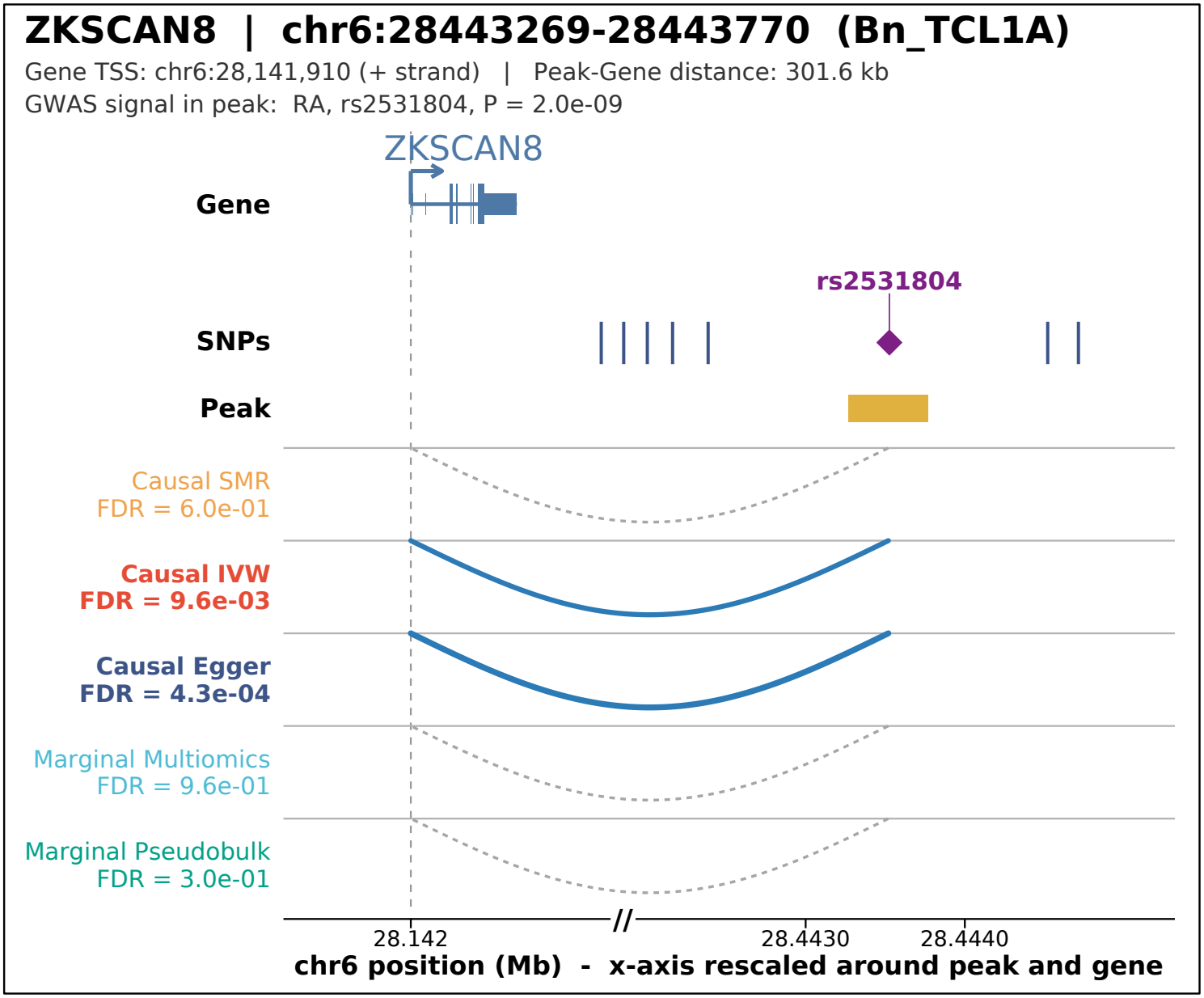

GLS2 | chr12:56041553-56042054 (CD8_Tn_CCR7)
Gene TSS: chr12:56,488,414 (- strand) | Peak-Gene distance: 446.6 kb
GWAS signal in peak: Asthma, rs705704, P = 1.9e-14 | T1D, rs705704, P = 1.5e-62
GLS2
Gene
rs705704
SNPs
Peak
Causal SMR
FDR = 1.4e-02
Causal IVW
FDR = 3.1e-04
Causal Egger
FDR = 5.8e-03
Marginal Multiomics
FDR = 4.7e-01
Marginal Pseudobulk
FDR = 3.2e-01
56.0410
56.0420
56.488
chr12 position (Mb) - x-axis rescaled around peak and gene

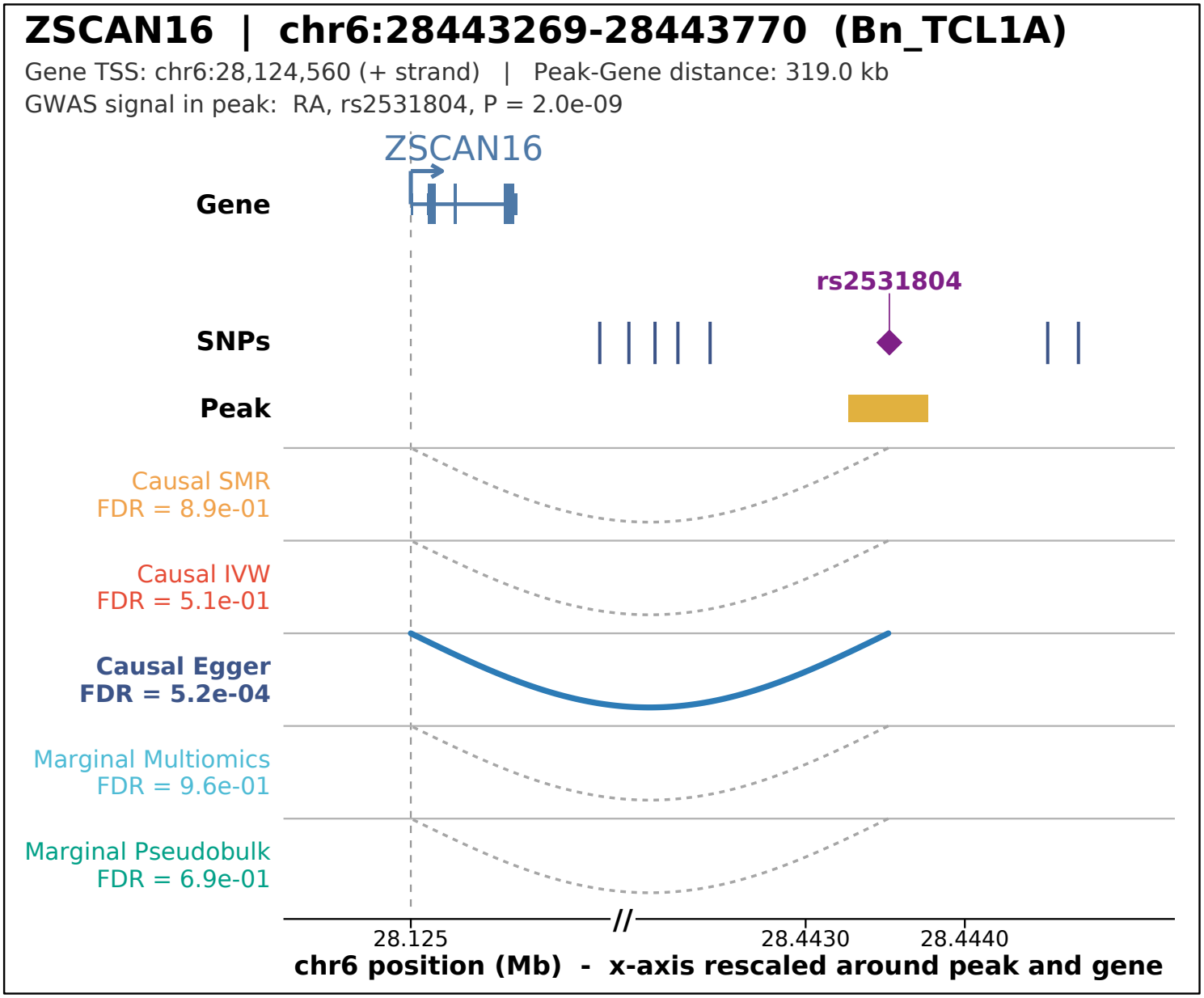

GNAS | chr20:59254001-59254502 (CD4_Tn_CCR7)
Gene TSS: chr20:58,839,718 (+ strand) | Peak-Gene distance: 414.5 kb
GWAS signal in peak: CD, rs259956, P = 8.4e-09 | IBD, rs259956, P = 2.6e-09
GNAS
Gene
rs259956
SNPs
Peak
Causal SMR
FDR = 3.4e-04
Causal IVW
FDR = 7.3e-03
Causal Egger
FDR = 4.4e-02
Marginal Multiomics
FDR = 5.9e-01
Marginal Pseudobulk
FDR = 5.2e-01
58.840
59.2535
59.2545
chr20 position (Mb) - x-axis rescaled around peak and gene

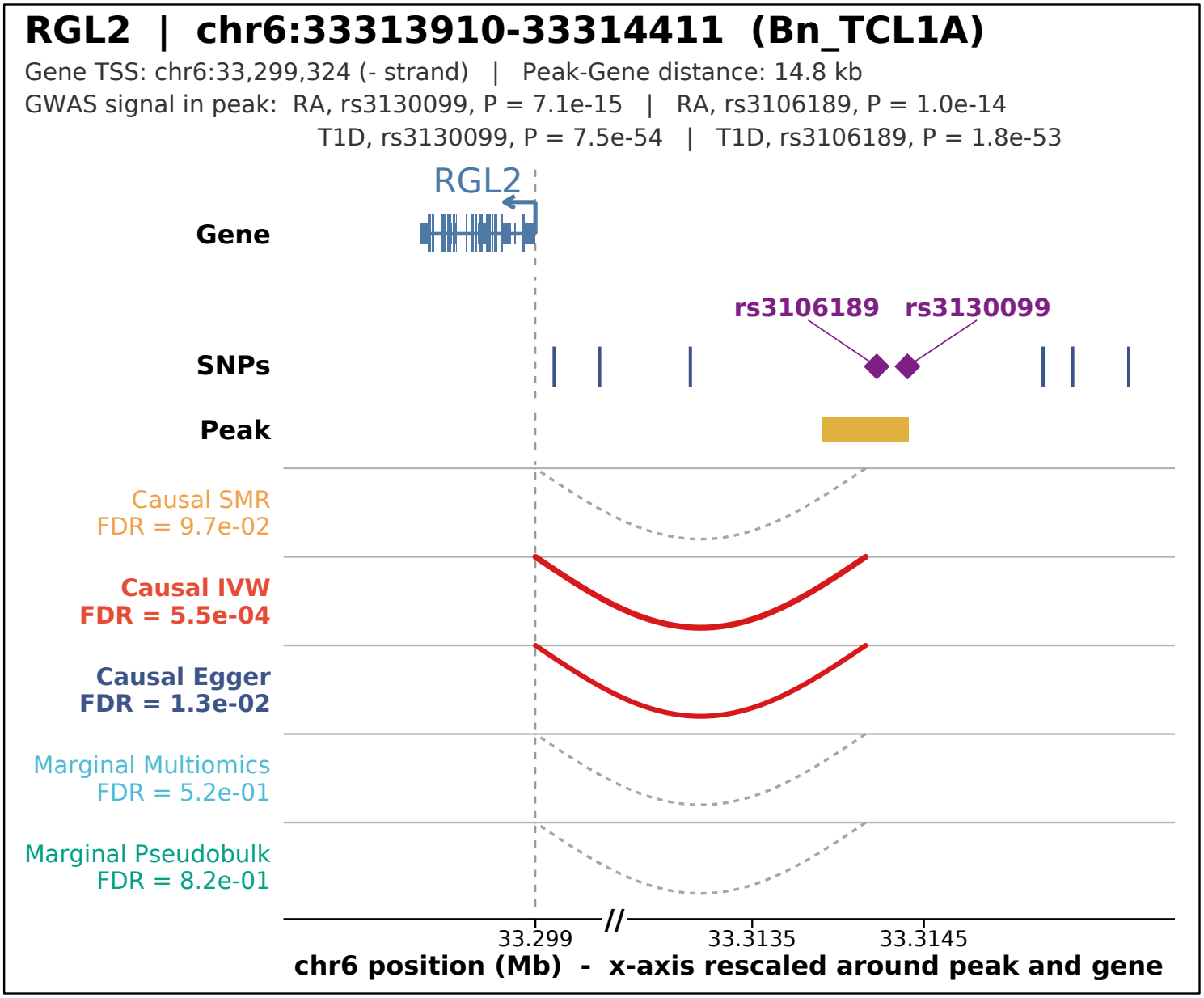

PLEKHH2 | chr2:43537114-43537615 (CD4_Tn_CCR7)
Gene TSS: chr2:43,637,273 (+ strand) | Peak-Gene distance: 99.9 kb
GWAS signal in peak: CD, rs17031016, P = 8.2e-10
PLEKHH2
Gene
rs17031016
SNPs
Peak
Causal SMR
FDR = 3.7e-04
Causal IVW
FDR = 9.5e-04
Causal Egger
FDR = 6.3e-01
Marginal Multiomics
FDR = 1.6e-01
Marginal Pseudobulk
FDR = 9.6e-02
43.5365
43.5375
43.637
chr2 position (Mb) - x-axis rescaled around peak and gene

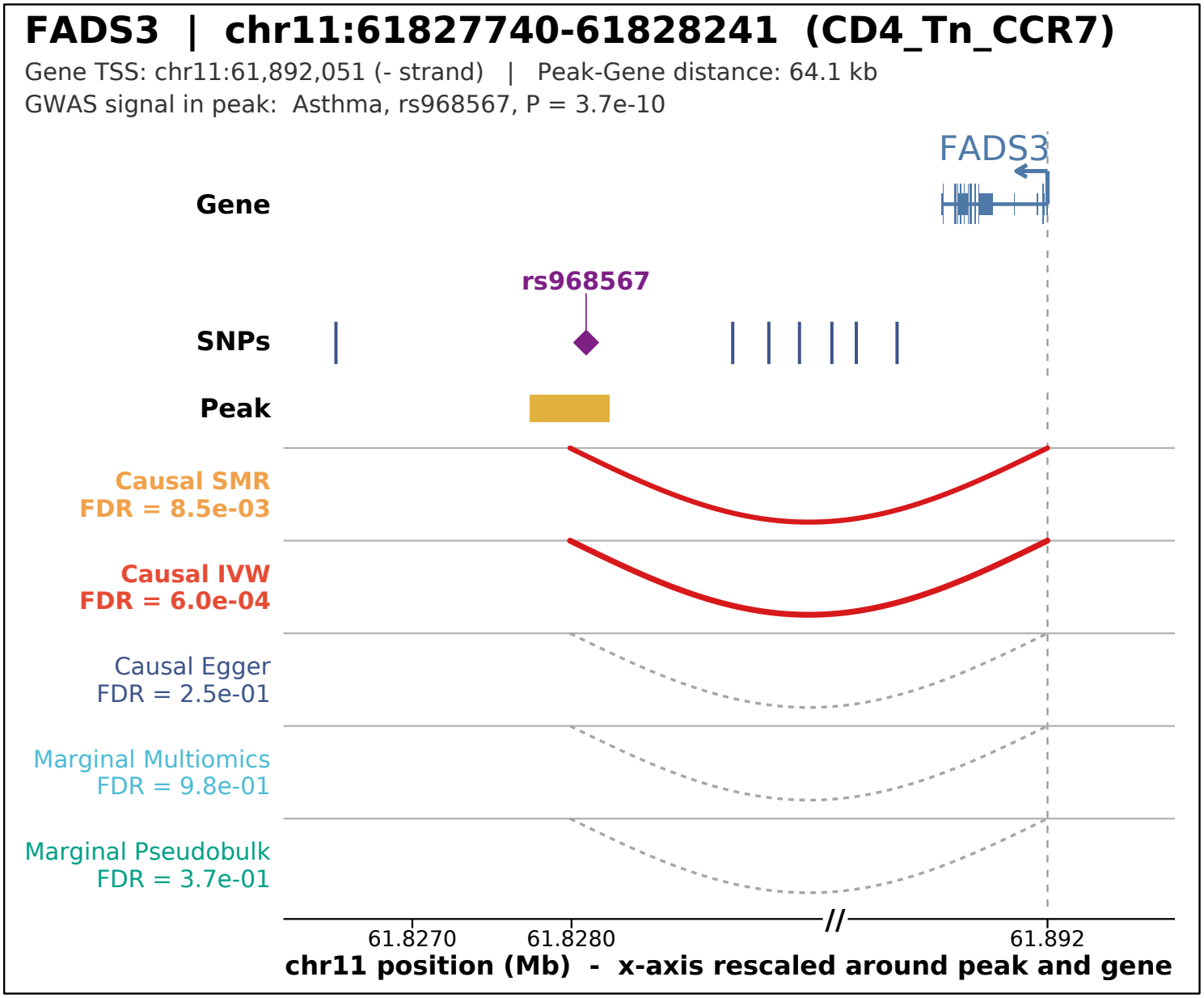

DR1 | chr1:93179440-93179941 (Bn_TCL1A)
Gene TSS: chr1:93,345,888 (+ strand) | Peak-Gene distance: 166.2 kb
GWAS signal in peak: CD, rs1335679, P = 3.9e-10
DR1
Gene
rs1335679
SNPs
Peak
Causal SMR
FDR = 5.8e-04
Causal IVW
FDR = 4.0e-04
Causal Egger
FDR = 1.8e-01
Marginal Multiomics
FDR = 9.4e-01
Marginal Pseudobulk
FDR = 3.2e-01
93.1790
93.1800
93.346
chr1 position (Mb) - x-axis rescaled around peak and gene


UBQLN4 | chr1:155227359-155227860 (CD8_Tn_CCR7)
Gene TSS: chr1:156,053,794 (- strand) | Peak-Gene distance: 826.2 kb
GWAS signal in peak: IBD, rs2990245, P = 7.2e-09
UBQLN4
Gene
rs2990245
SNPs
Peak
Causal SMR
FDR = 7.5e-01
Causal IVW
FDR = 3.0e-03
Causal Egger
FDR = 4.2e-04
Marginal Multiomics
FDR = 3.9e-01
Marginal Pseudobulk
FDR = 3.8e-01
155.2270
155.2280
156.054
chr1 position (Mb) - x-axis rescaled around peak and gene


ZKSCAN8 | chr6:28443269-28443770 (Bn_TCL1A)
Gene TSS: chr6:28,141,910 (+ strand) | Peak-Gene distance: 301.6 kb
GWAS signal in peak: RA, rs2531804, P = 2.0e-09
ZKSCAN8
Gene
rs2531804
SNPs
Peak
Causal SMR
FDR = 6.0e-01
Causal IVW
FDR = 9.6e-03
Causal Egger
FDR = 4.3e-04
Marginal Multiomics
FDR = 9.6e-01
Marginal Pseudobulk
FDR = 3.0e-01
28.142
28.4430
28.4440
chr6 position (Mb) - x-axis rescaled around peak and gene


ZSCAN16 | chr6:28443269-28443770 (Bn_TCL1A)
Gene TSS: chr6:28,124,560 (+ strand) | Peak-Gene distance: 319.0 kb
GWAS signal in peak: RA, rs2531804, P = 2.0e-09
ZSCAN16
Gene
rs2531804
SNPs
Peak
Causal SMR
FDR = 8.9e-01
Causal IVW
FDR = 5.1e-01
Causal Egger
FDR = 5.2e-04
Marginal Multiomics
FDR = 9.6e-01
Marginal Pseudobulk
FDR = 6.9e-01
28.125
28.4430
28.4440
chr6 position (Mb) - x-axis rescaled around peak and gene


RGL2 | chr6:33313910-33314411 (Bn_TCL1A)
Gene TSS: chr6:33,299,324 (- strand) | Peak-Gene distance: 14.8 kb
GWAS signal in peak: RA, rs3130099, P = 7.1e-15 | RA, rs3106189, P = 1.0e-14
T1D, rs3130099, P = 7.5e-54 | T1D, rs3106189, P = 1.8e-53
RGL2
Gene
rs3106189
rs3130099
SNPs
Peak
Causal SMR
FDR = 9.7e-02
Causal IVW
FDR = 5.5e-04
Causal Egger
FDR = 1.3e-02
Marginal Multiomics
FDR = 5.2e-01
Marginal Pseudobulk
FDR = 8.2e-01
33.299
33.3135
33.3145
chr6 position (Mb) - x-axis rescaled around peak and gene


FADS3 | chr11:61827740-61828241 (CD4_Tn_CCR7)
Gene TSS: chr11:61,892,051 (- strand) | Peak-Gene distance: 64.1 kb
GWAS signal in peak: Asthma, rs968567, P = 3.7e-10
FADS3
Gene
rs968567
SNPs
Peak
Causal SMR
FDR = 8.5e-03
Causal IVW
FDR = 6.0e-04
Causal Egger
FDR = 2.5e-01
Marginal Multiomics
FDR = 9.8e-01
Marginal Pseudobulk
FDR = 3.7e-01
61.8270
61.8280
61.892
chr11 position (Mb) - x-axis rescaled around peak and gene